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The Memory and CPU Use of Processes in Windows

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[redacted]

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Actual Word Count: 3267

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Module: Operating Systems
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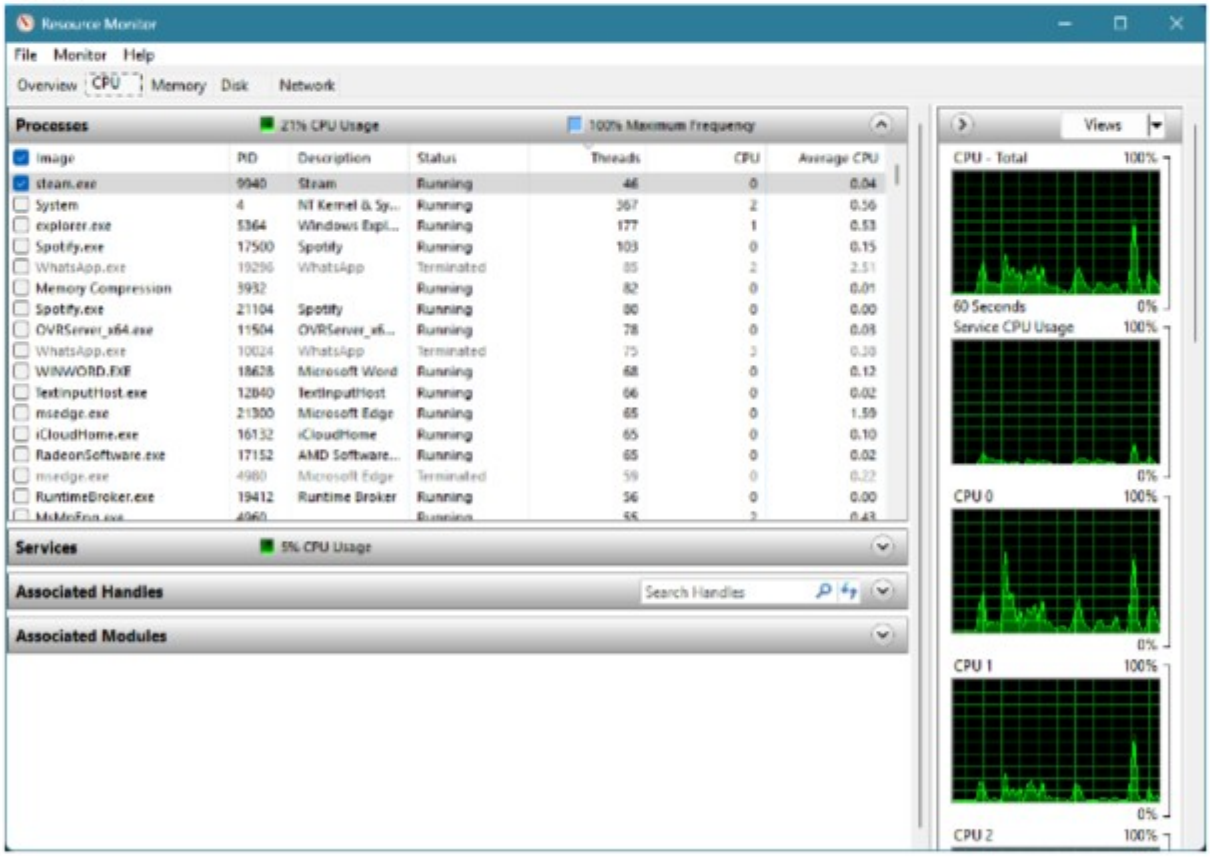
Steam

Steam is the largest videogame library as of today, it has all kinds of games for Windows computers, Macs & MacBooks, and Steam Decks. The process is called 'steam.exe' and has 9940 as its Process ID.

1. *Browsing the Library*

When browsing the games store and my own library, there isn't much CPU consumption at all. The average percent of CPU consumption (every 60 seconds) whether I'm going through the games' store pages or my library is between 0.02% and 0.05%. However, when going through the pictures and watching the trailers of some games, it can jump up to 0.06% or 0.07%. The active thread count consistently sits at around 41-43 and can sometimes go down to 39 or jump up to 46. The CPU spikes shown in the graph are likely due to the videos being played on the Steam app.

CPU:

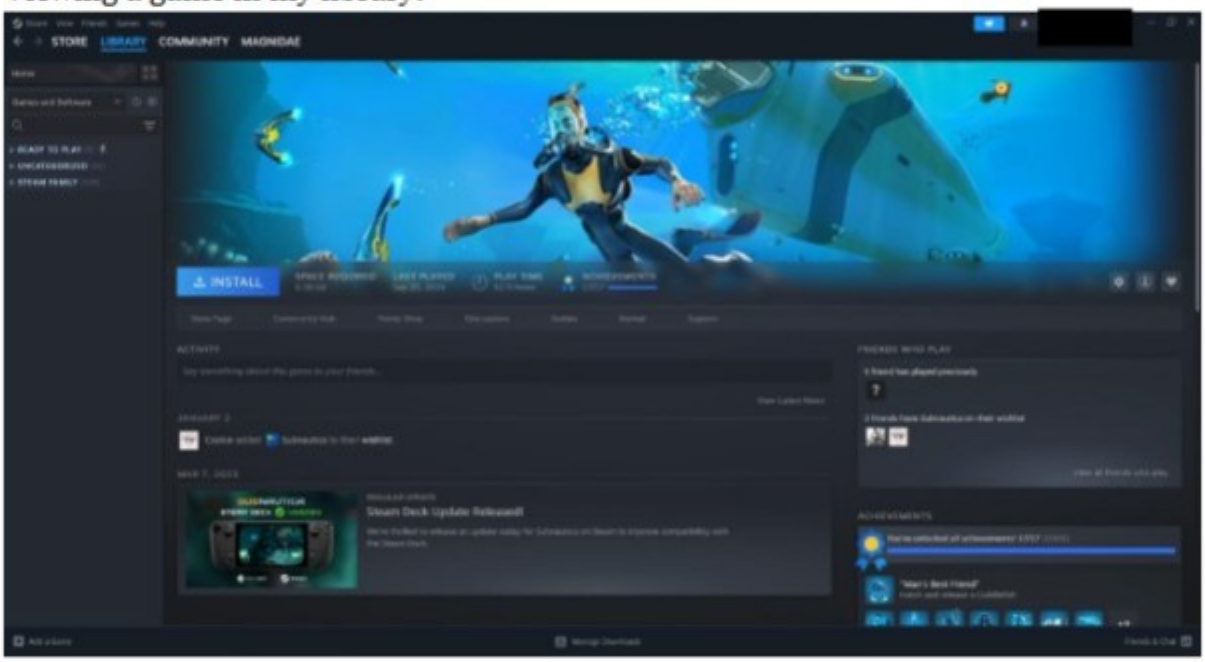


As for the memory, it consistently stays at around 68k KB for Commit, 52k KB for Working Set, 34k KB for Shareable, and 18k KB for Private.

Viewing a game's store page:



Viewing a game in my library:



2. *Downloading a Game*

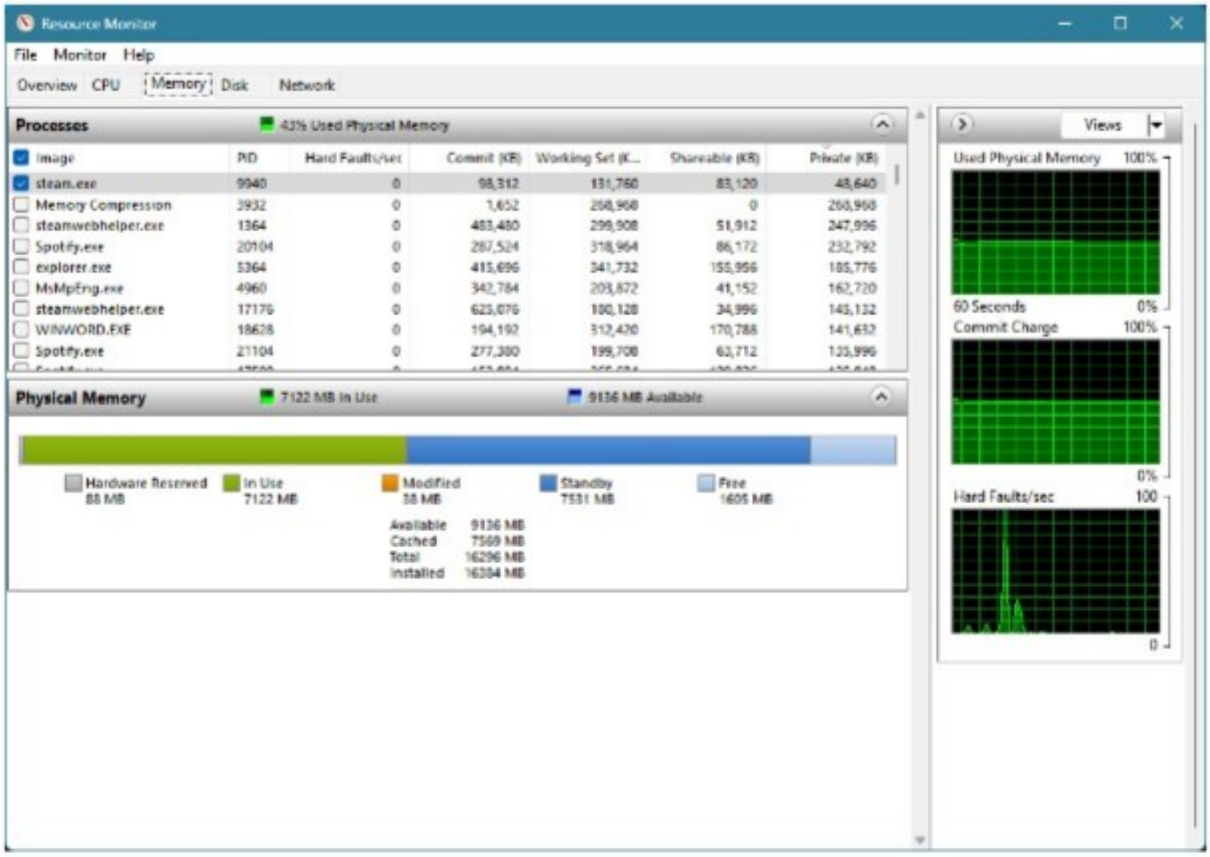
Downloading a game does demand more resources than simply browsing them. I took the example of Subnautica, a nice game that needs only 6.58GB to download (which in today's standards is quite low).

When downloading this game, the active thread count peaks at 86, with the current CPU usage being at 4%, and the average CPU usage peaking at around 2.7%.

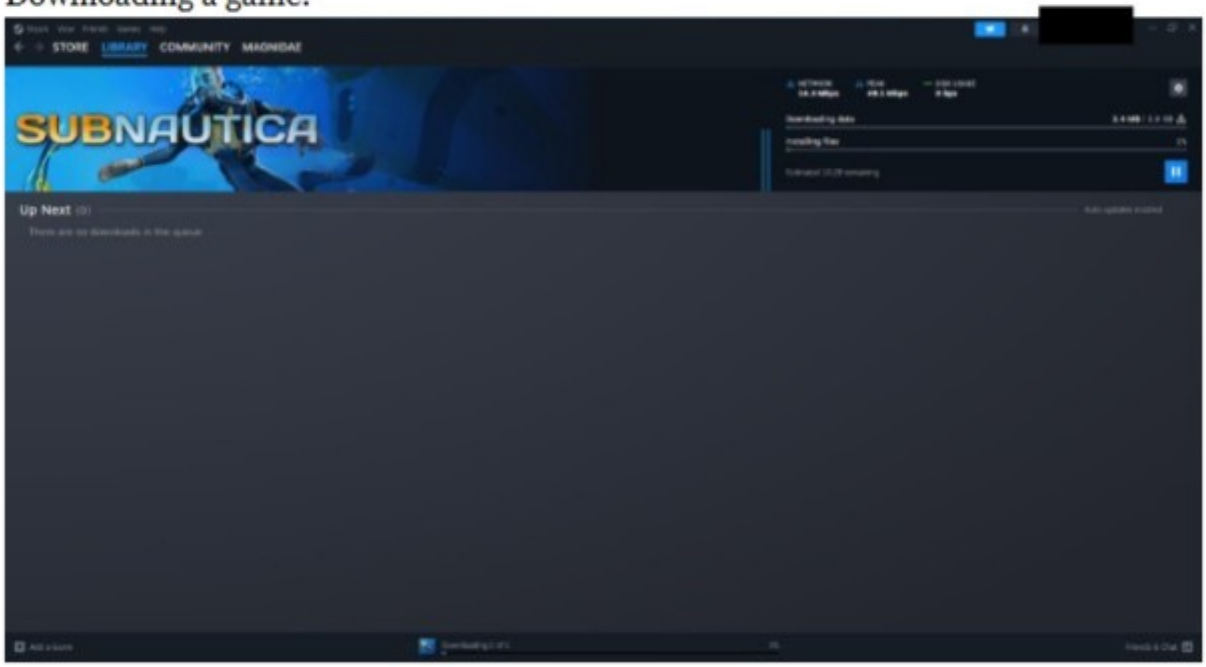
For the memory use, it reaches heights such as 153k KB for Commit, 164k KB for Working Set, 83k KB for Shareable, and 90k KB for Private. These values increase over time as the game is being downloaded. Then, once the game has been downloaded (or

the download has been cancelled by the user), everything decreases a bit (except for the Shareable).

Memory:



Downloading a game:



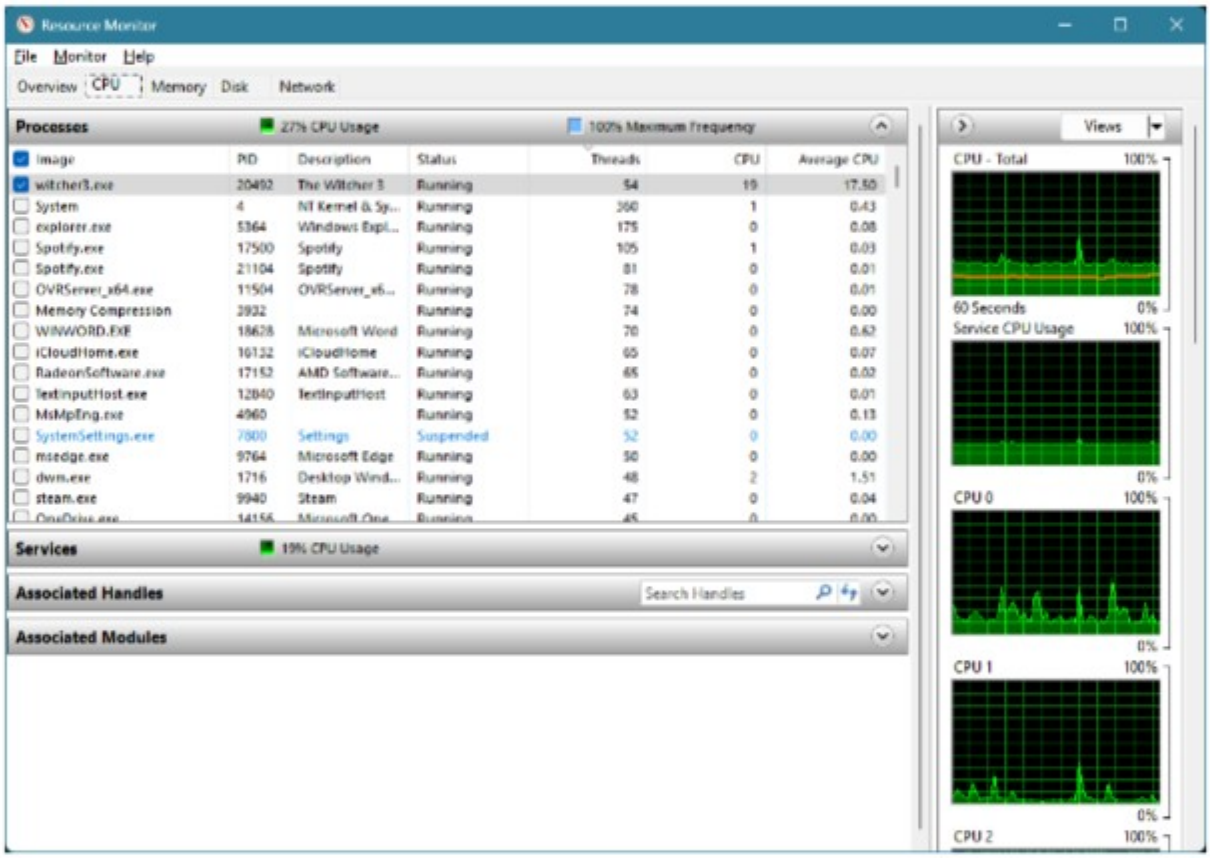
The Witcher 3: Wild Hunt

The Witcher 3: Wild Hunt is a very popular open-world RPG game based of the books by Andrzej Sapkowski. It is very demanding game graphically (which made me wonder what impact it had on the CPU mostly), especially with Ray Tracing enabled. The process is called 'witcher3.exe' and has 20492 as its Process ID.

1. *The Game Menu*

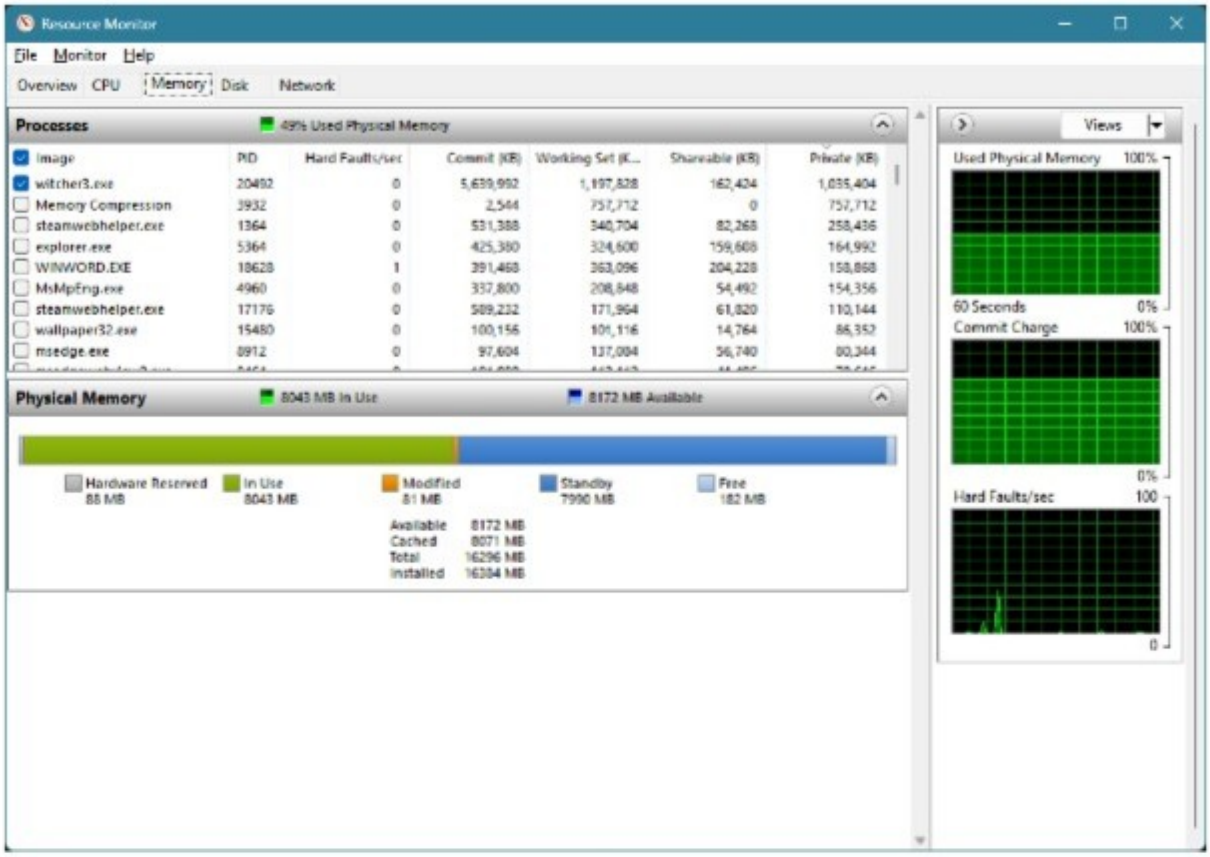
When in the game menu, its active thread count peaks at 62 and can go down to 55, while its current CPU usage is already at 17-18%, and its average usage peaking at around 17.4% (decreasing to around 16.5%). This is impressive considering the game isn't technically being played yet. However, it is somewhat expected: the game menu plays music and renders a HD video.

CPU:



As for the memory usage, when in the menu, around 5.6 million KB are for Commit, 1.2 million KB for Working Set, 163k KB for Shareable, and 1 million KB for Private. Which is a very big leap from the previous testing on Steam.

Memory:



Game menu:

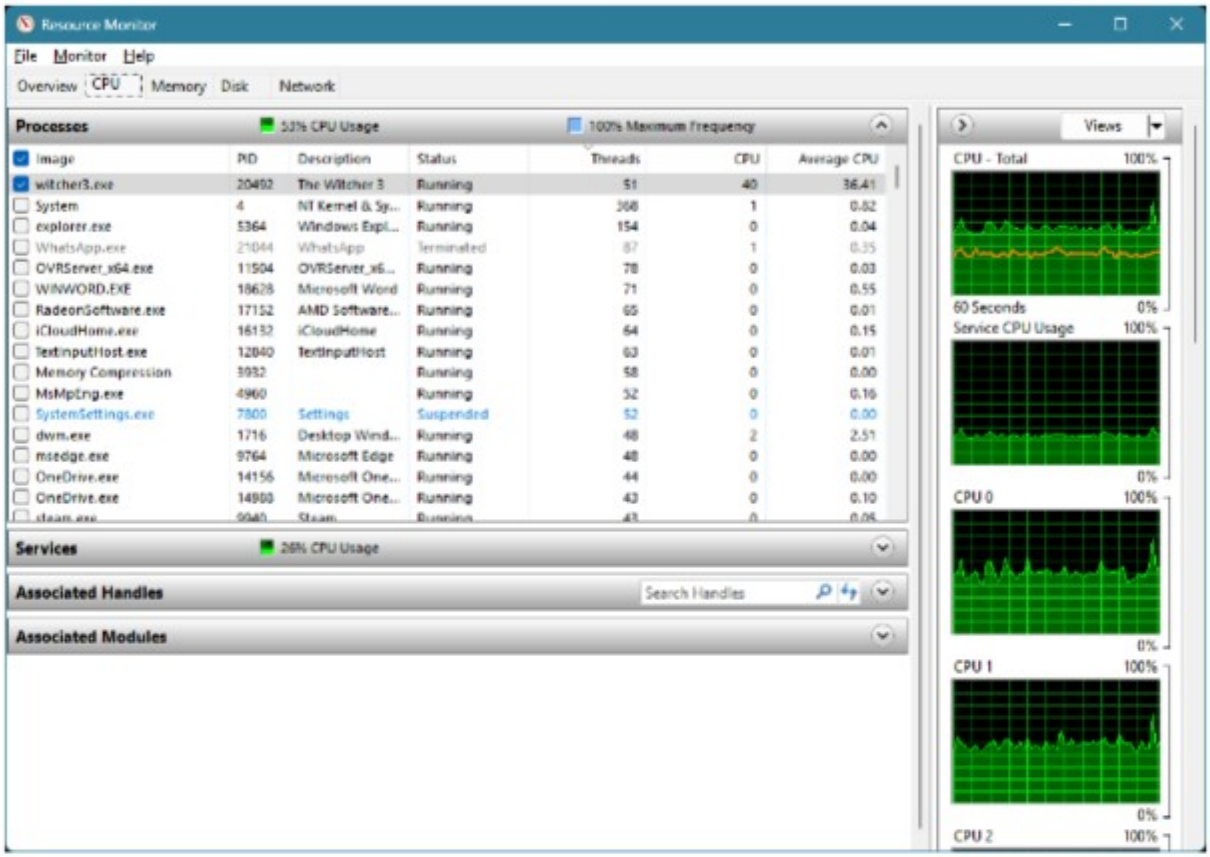


2. *Playing the Game*

When playing the game, everything jumps much higher because this game is very demanding for all kinds of resources. In addition, I am playing the game at the highest possible settings (excluding Ray Tracing, it is currently off). It has to render so many

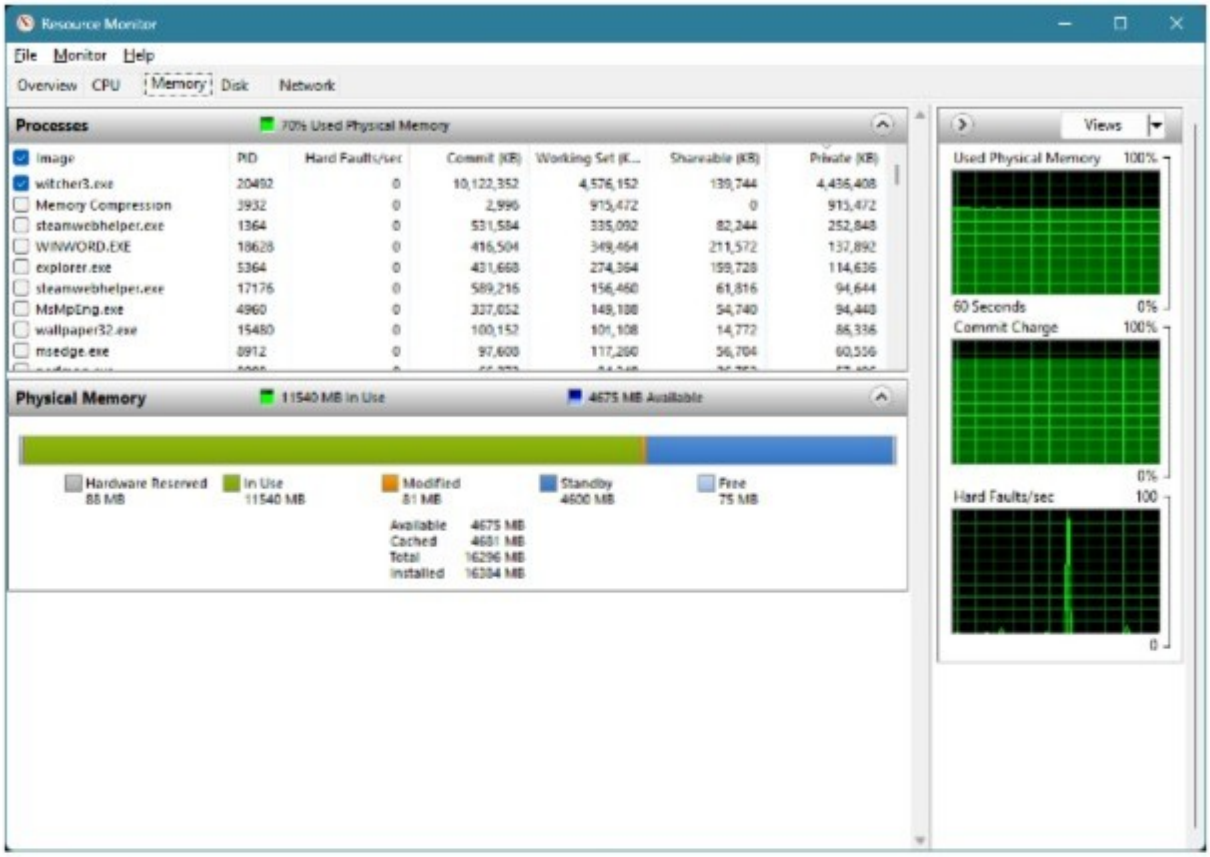
objects and details, and has to handle game mechanics and user inputs at the same time. For the CPU, the active thread count stays rather similar with the menu, currently sitting at 58, and can decrease down to 51. The current CPU usage goes up to 40%, with the average staying at around 36-37%, which is very high.

CPU:



For the memory, it is in much higher use than in the menu (except for the Shareable), 10 million KB are for Commit, 4.6 million KB for Working Set, 140k KB for Shareable, and 4.5 million KB for Private. There were even 2 hard faults per second at times, and the graph below also shows that the hard faults per second peaked over 90.

Memory:



Playing the game:

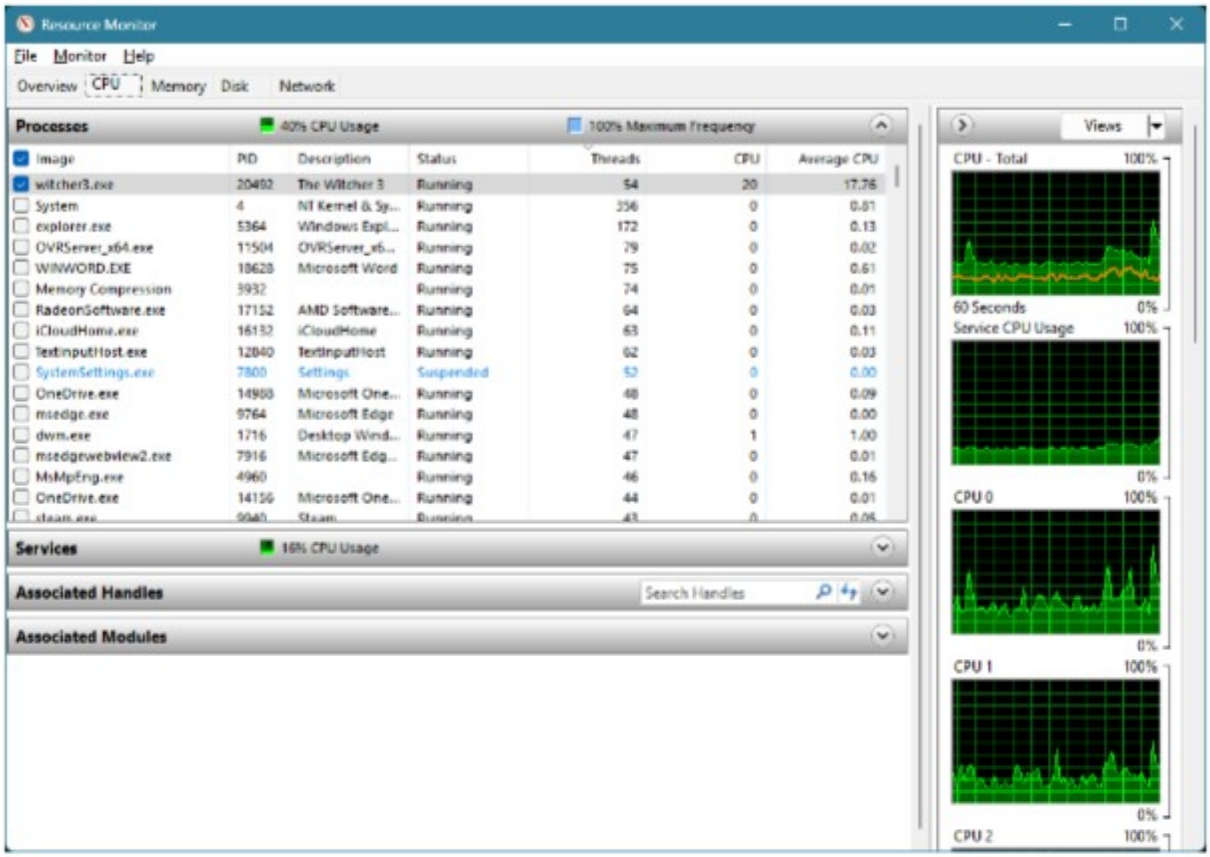


3. *Modifying Parameters (activating Ray Tracing)*

Ray Tracing increases the GPU consumption by a lot. But how does this compare to the CPU and memory consumptions? With Ray Tracing on and set on the highest settings, the active thread count stays pretty much the same, and surprisingly, the CPU

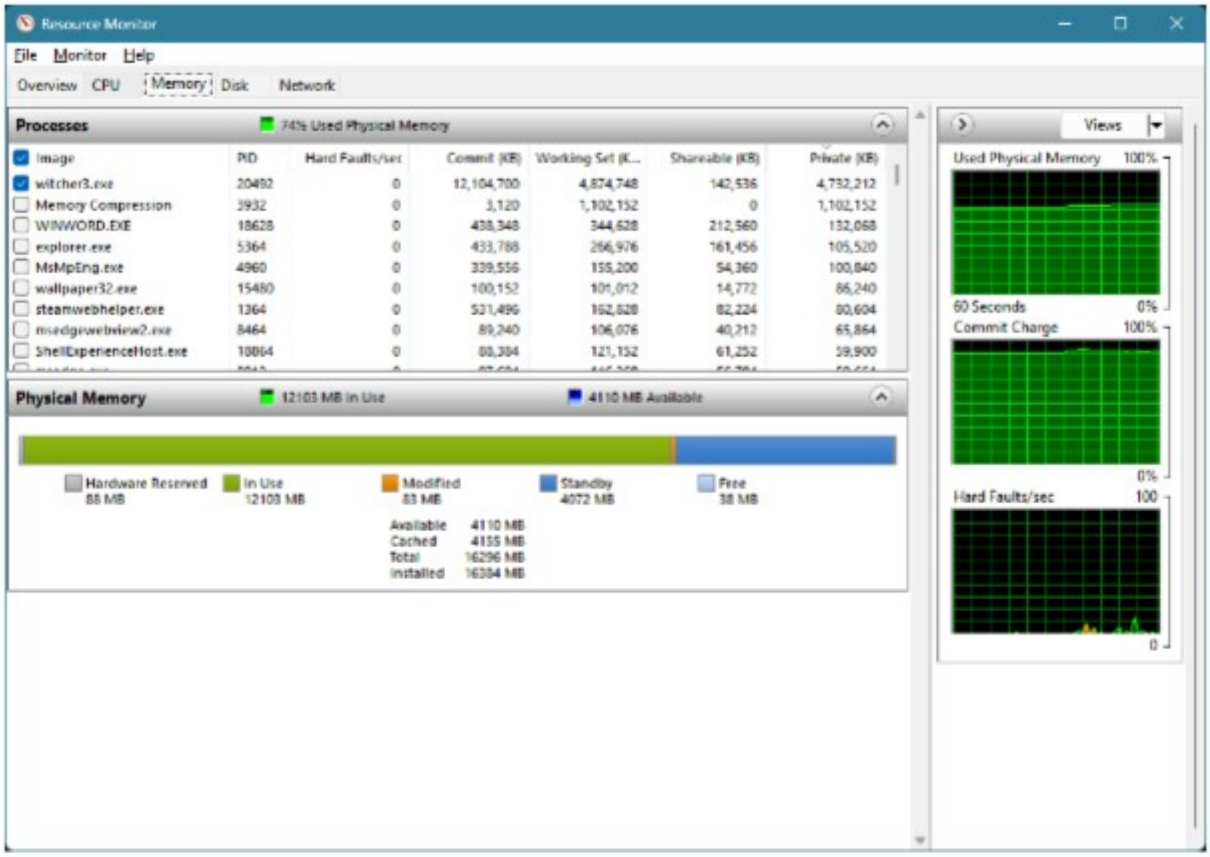
peaks at a current use of 20%, and then lingers at around 15-17%, which is much lower than the previous test (and the average CPU usage stays at around 16% too). This is all because, in general when activating Ray Tracing, the GPU takes on more of the workload than before (especially for graphical processing), so less is left for the CPU.

CPU:



For the memory, it is also practically the exact same, with slight increases (such as a 2 million KB increase for Commit, and roughly a 200k KB increase for both Working Set and Private).

Memory:



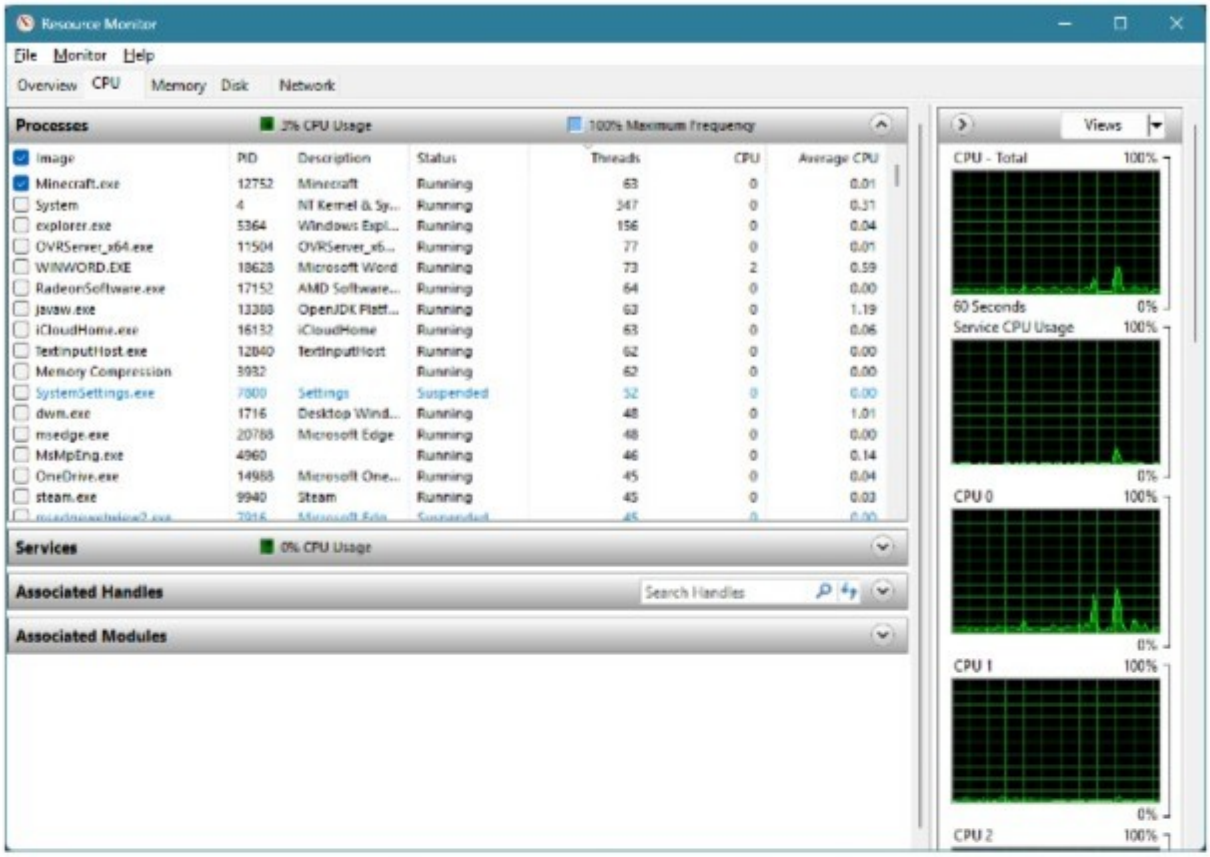
Minecraft: Java Edition

Minecraft is one of the world's most popular games (if not the most popular one) that places the player in an infinite, open-world, blocky survival and creative game. Some relatively famous fact about Minecraft Java Edition is that despite it being the preferred version of Minecraft for most users, it is terribly optimised: for example, it renders everything, even what the player isn't directly looking at, which is a waste of resources being used. The process name is 'Minecraft.exe', and has 12752 as its Process ID.

1. *The Game Menu*

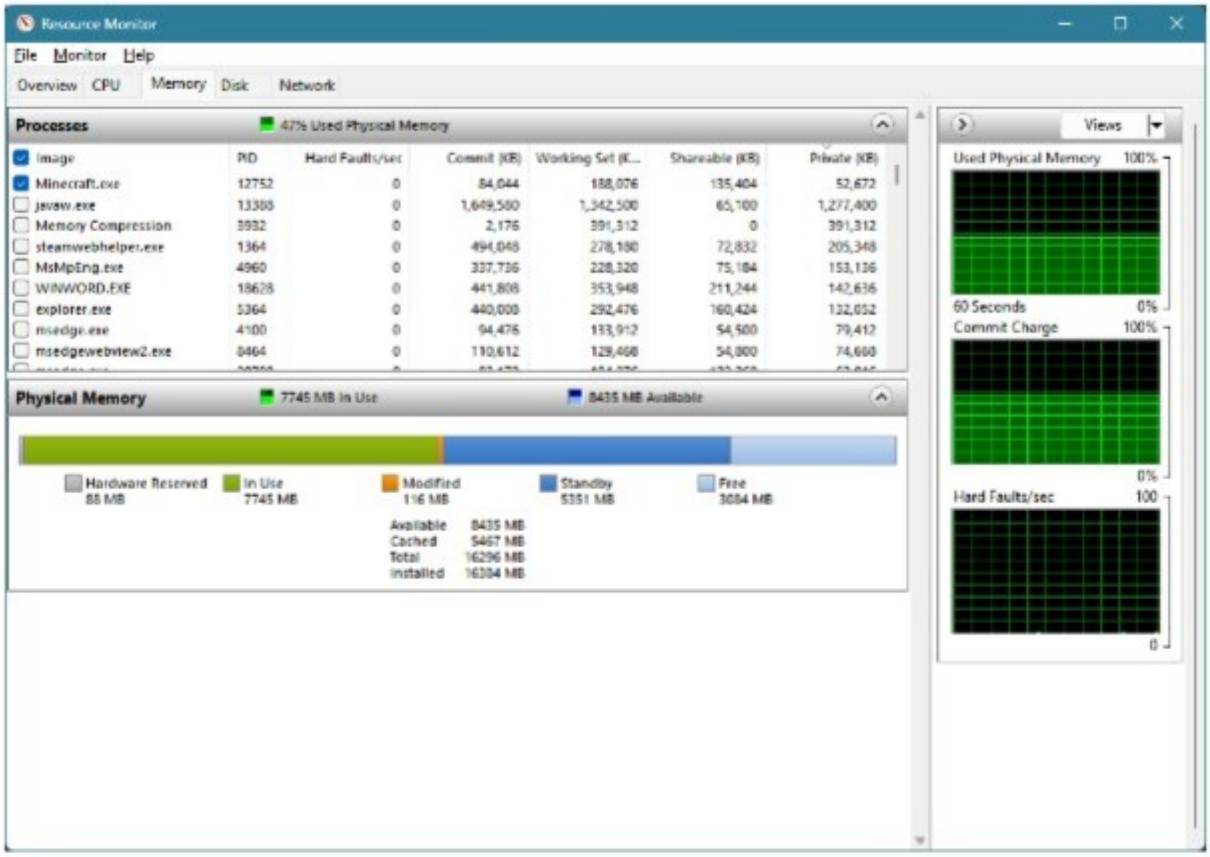
The game menu is simple, it plays music (like The Witcher), and displays a moving background image (instead of a video). The CPU isn't used much at all, quite similar to Steam, with 63 active threads, and only 0.02% of average CPU consumption.

CPU:



For the memory, it is also quite low, or at least much lower than The Witcher. Roughly 84k KB are for Commit, 188k KB for Working Set, 125k KB for Shareable, and 53k for Private.

Memory:



Game menu:

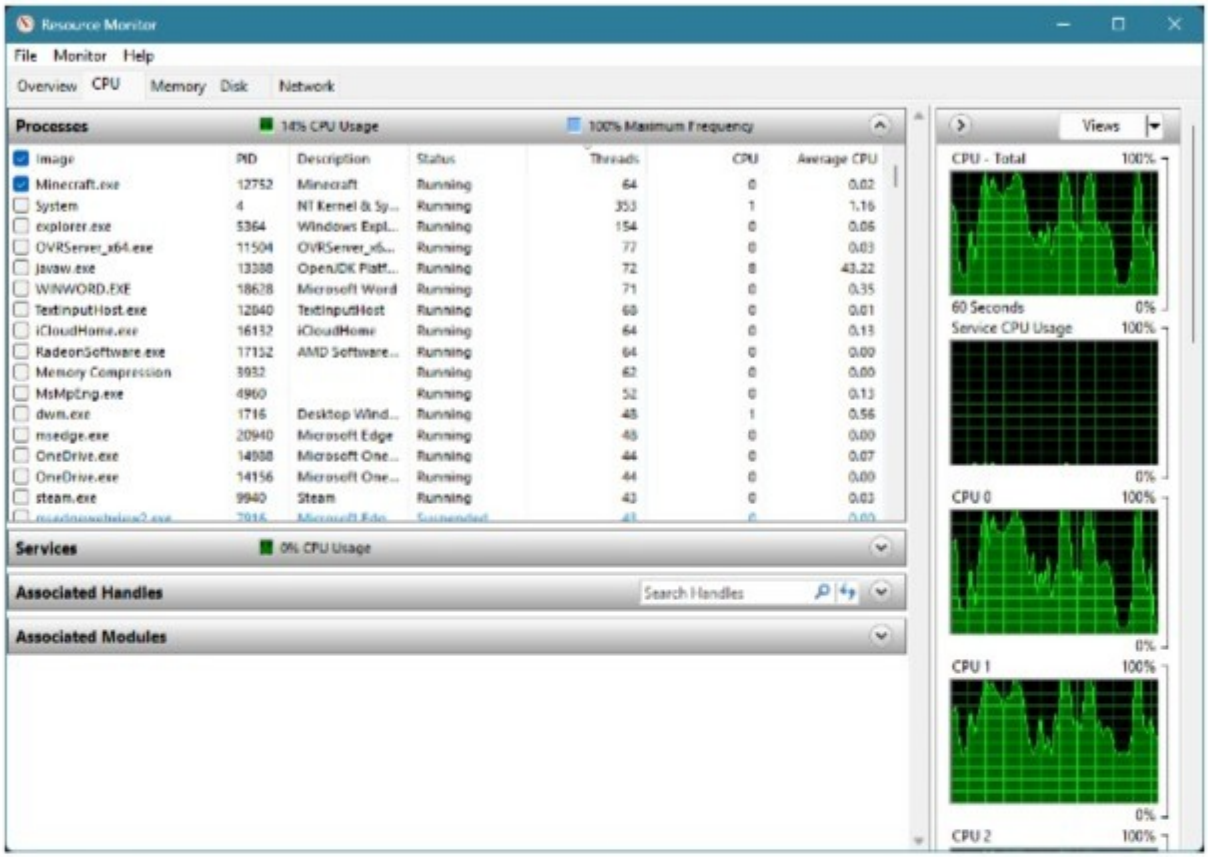


2. *Playing the Game*

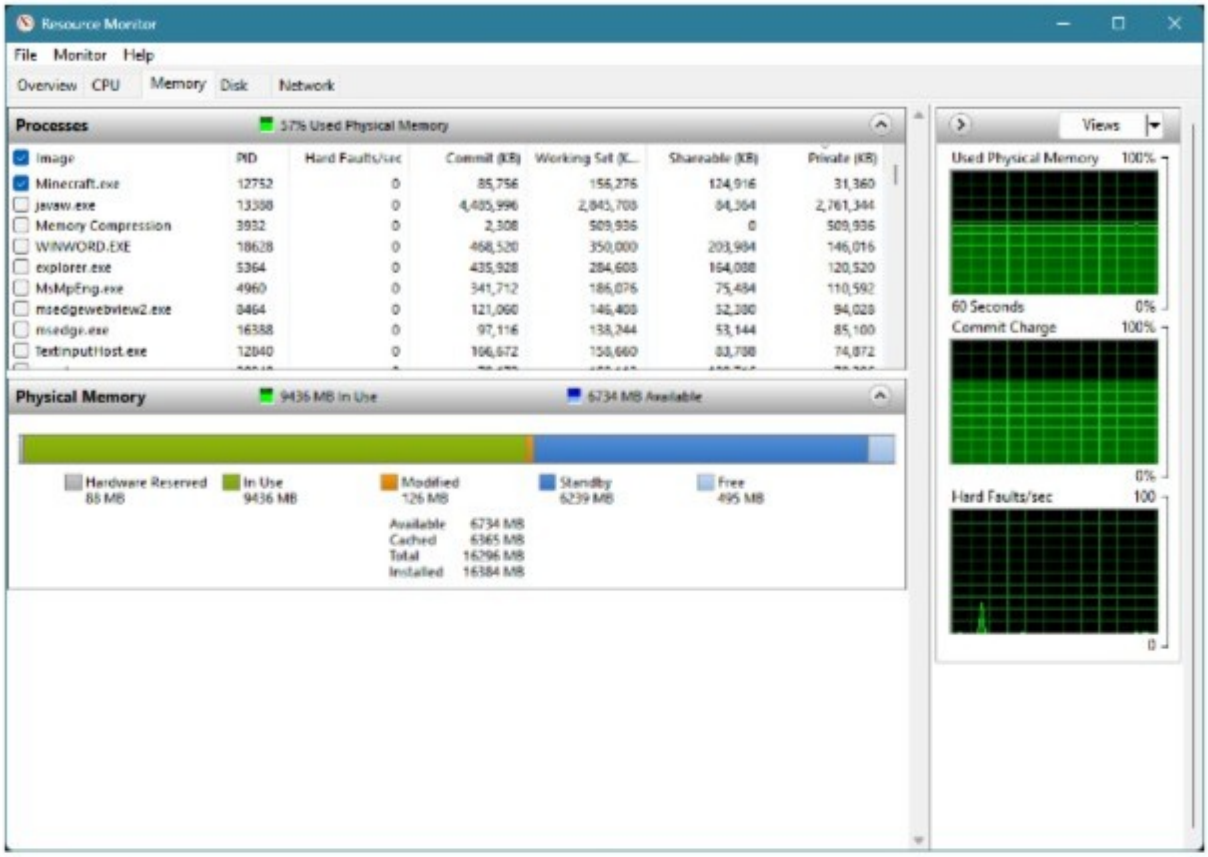
Playing the game in singleplayer mode involves creating an entire world for myself and playing inside of it, interacting with everything. But since my PC has a pretty powerful dedicated GPU, the app fully relies on it, leading to very minimal changes with the CPU

consumption compared to the menu when playing the game (the only main difference being the average CPU consumption rarely rising to 0.05%). And quite surprisingly, the memory also has zero changes, if anything, the overall memory in use has decreased.

CPU:



Memory:



Playing the game:



Observation

It is clear that games that require more resources to render elements and to process actions (such as user inputs or characters' AI) will inevitably be leading to more CPU consumption (as demonstrated in the Witcher 3) and memory usage.

Mark's Processes – [redacted]

I have chosen to analyse osu!(Lazer), Discord, and Apex Legends because I use them frequently. I expect Discord will need the least CPU and memory resources. Meanwhile, Apex Legends will require the most. And osu will sit somewhere in the middle.

The specifications of my PC are:
Processor: 11th Gen Intel(R) Core(TM) i9-11900KF @ 3.50GHz
RAM: 64GB DDR4 3333 MHz
Windows Version: 11 Pro (23H2)

The only monitoring software that I have used was Resource Monitor.

osu!(Lazer)

osu! is a rhythm game based on the gameplay of a variety of popular commercial rhythm games such as Osu! Tatakae! Ouendan and Elite Beat Agents. (Reddit 2024)

1. *Main menu (Idle)*



Figure 1: osu!(Lazer) main menu

When sitting in the main menu, the CPU consumption is very low. The average percent of CPU consumption (every 60 seconds) is 0.01%.

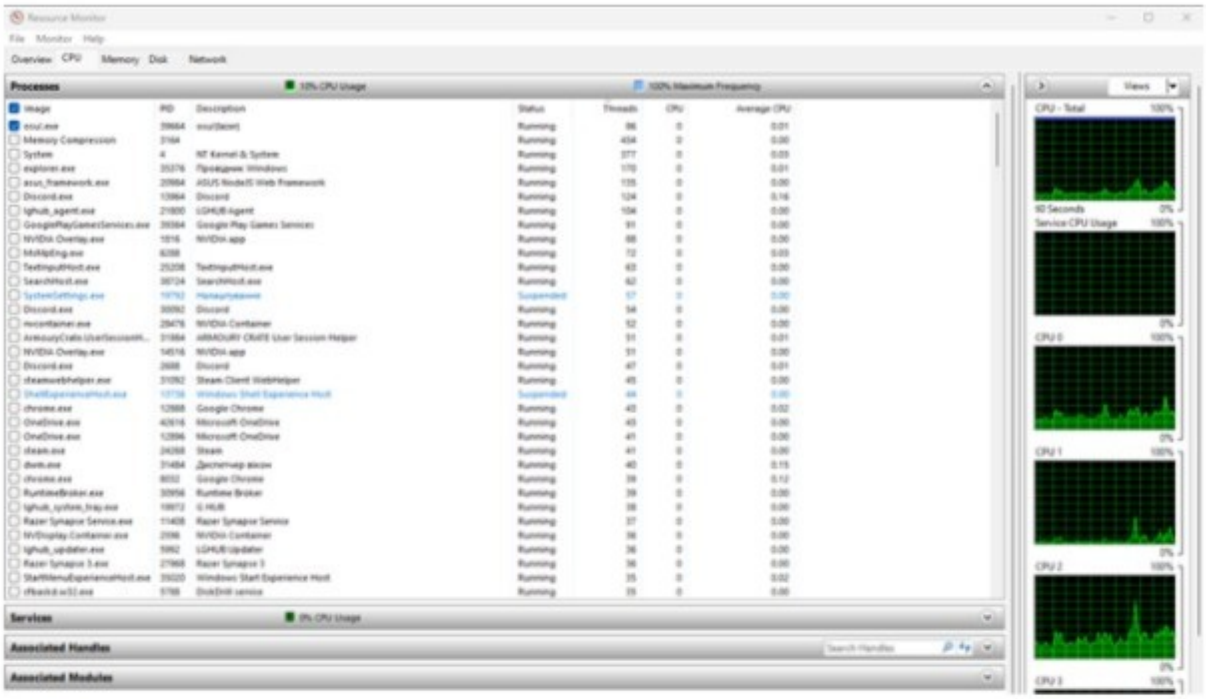


Figure 2: osu!(Lazer) main menu CPU consumption

As for the memory, it stays at around 327k KB for Commit, 419k KB for Working Set, 220k KB for Shareable, and 199k KB for Private.

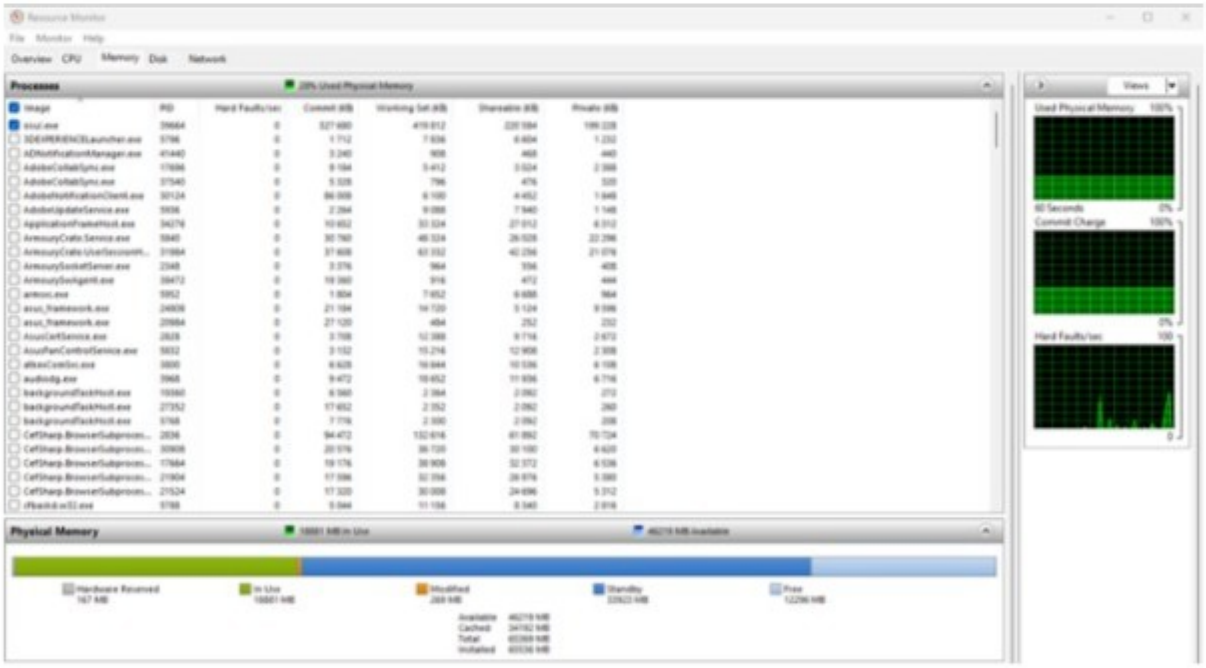


Figure 3: osu!(Lazer) main menu memory consumption

2. *Song select menu*

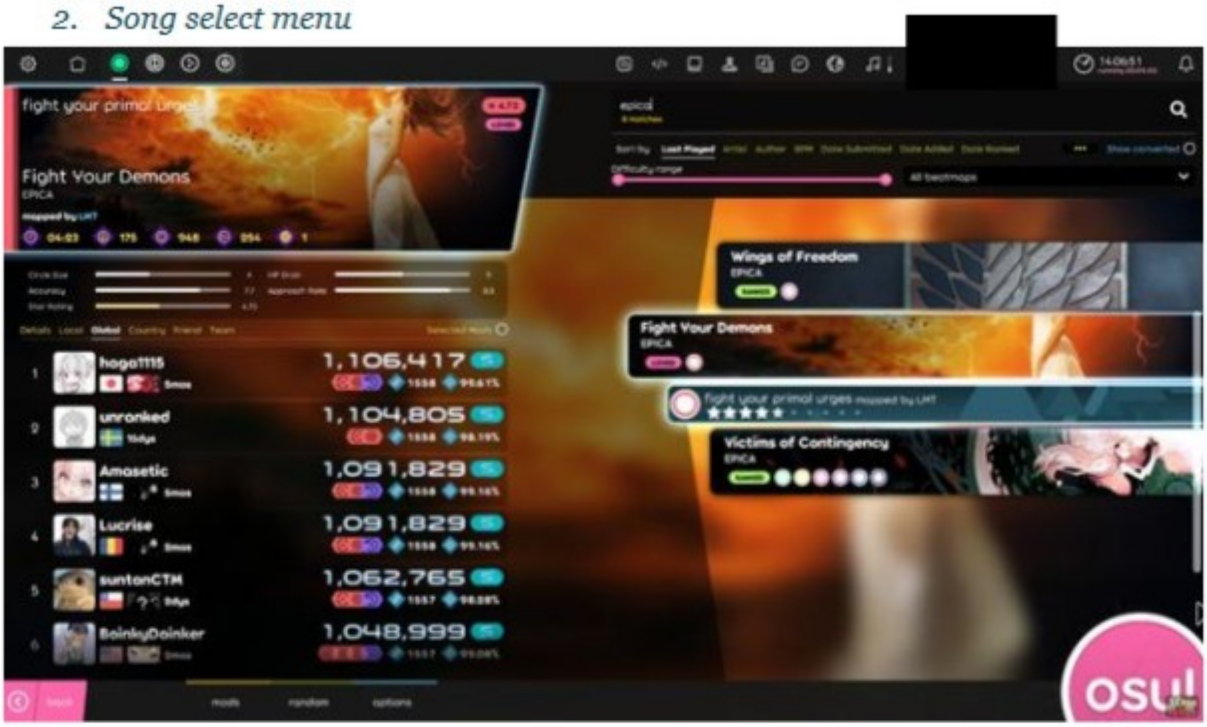


Figure 4: osu!(Lazer) song select menu

Navigating to the song selection menu. The CPU consumption grows quite a bit. From 0.01% to 0.72%.

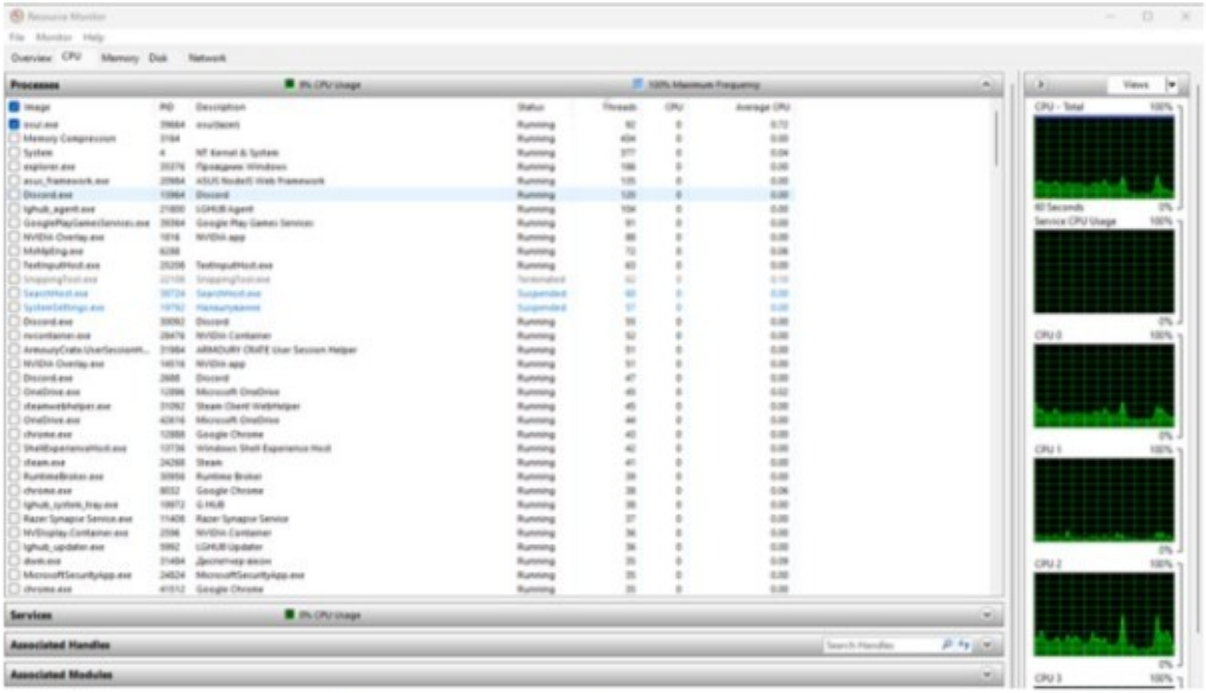


Figure 5: osu!(Lazer) song select menu CPU consumption

When looking at the memory consumption. It grows to 880k KB for Commit, 698k KB for Working Set, 252k KB for Shareable, and 446k KB for Private.

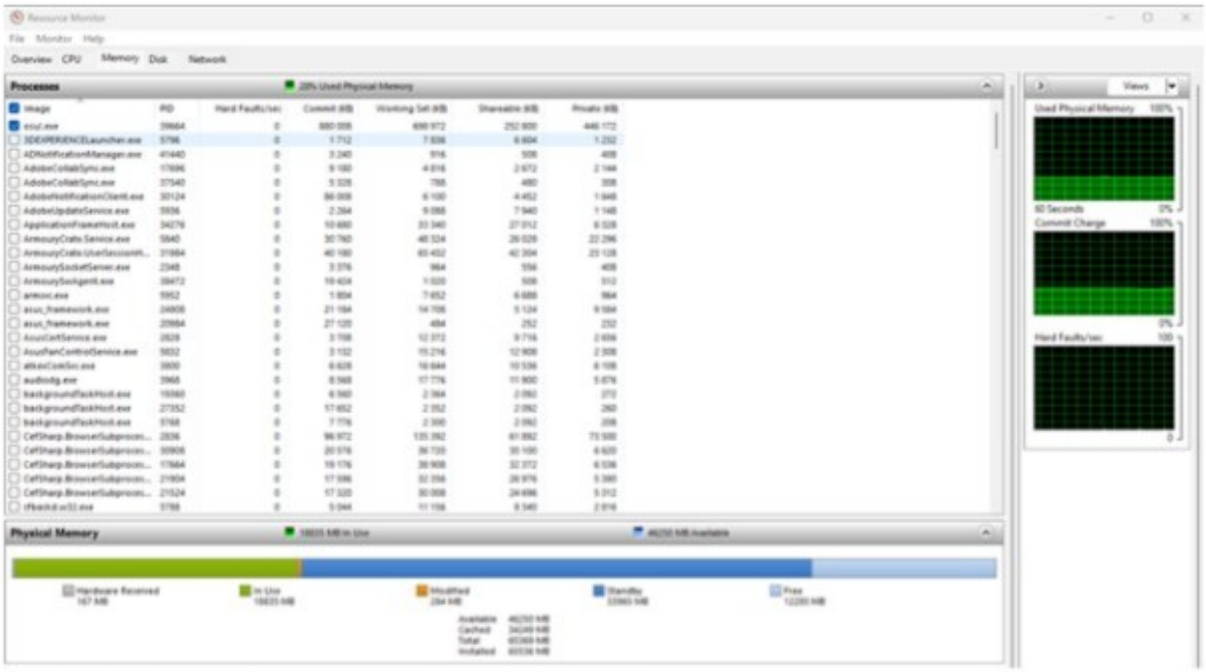


Figure 6: osu!(Lazer) song select menu memory consumption

3. *Playing a map*



Figure 7: osu!(Lazer) gameplay

While playing a map, the CPU consumption goes up again. It goes from 0.72% to 1.12%.

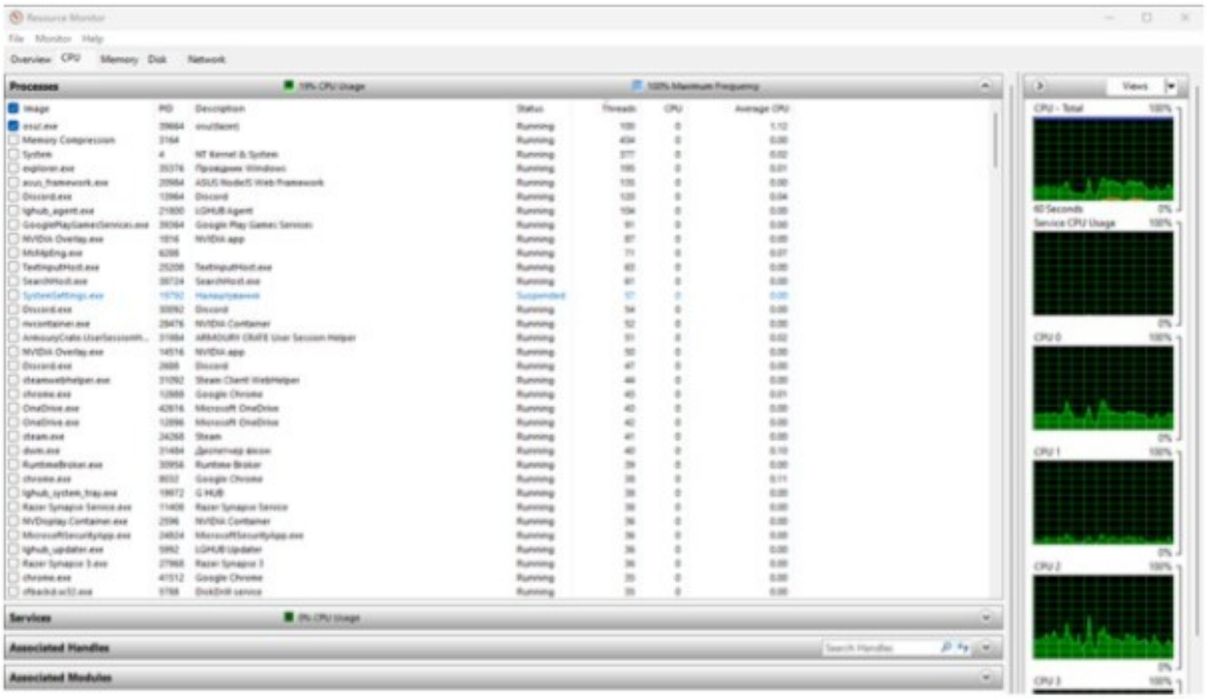


Figure 8: osu!(Lazer) gameplay CPU consumption

Looking at the memory consumption, it climbs up from the song selection menu. It is now 1.2m KB for Commit, 748k KB for Working Set, 247k KB Shareable, and 500k KB for Private.

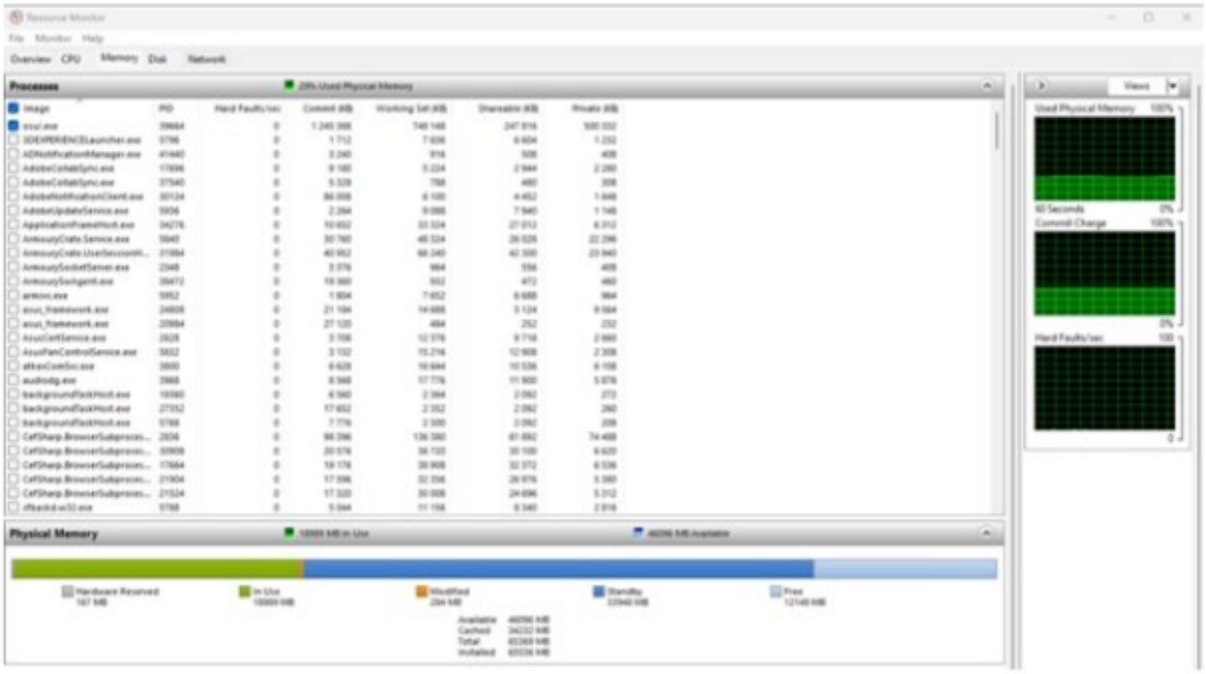


Figure 9: osu!(Lazer) gameplay memory consumption

Discord

Discord is an instant messaging and VoIP social platform which allows communication through voice calls, video calls, text messaging, and media. (Wikipedia 2024a)

1. *Server screen (idle)*

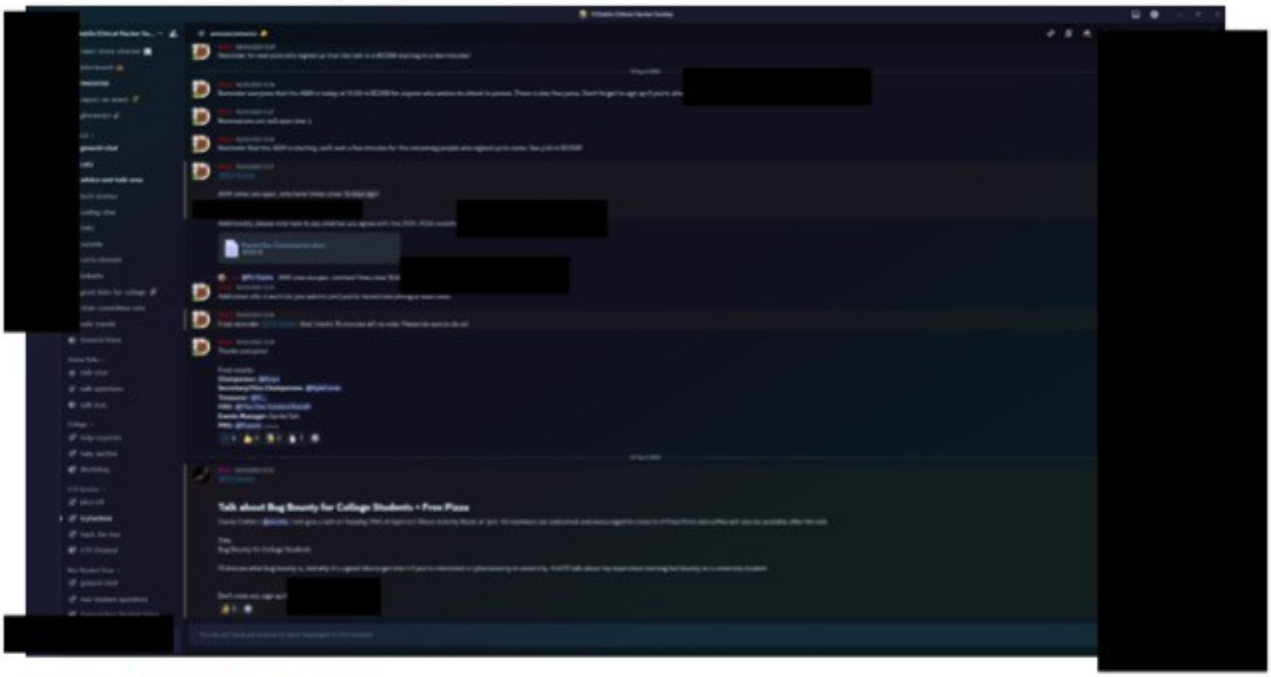


Figure 10: Discord server screen

While looking at the server screen, the CPU consumption is fairly low. The average percent of CPU consumption (every 60 seconds) is 0.22%.

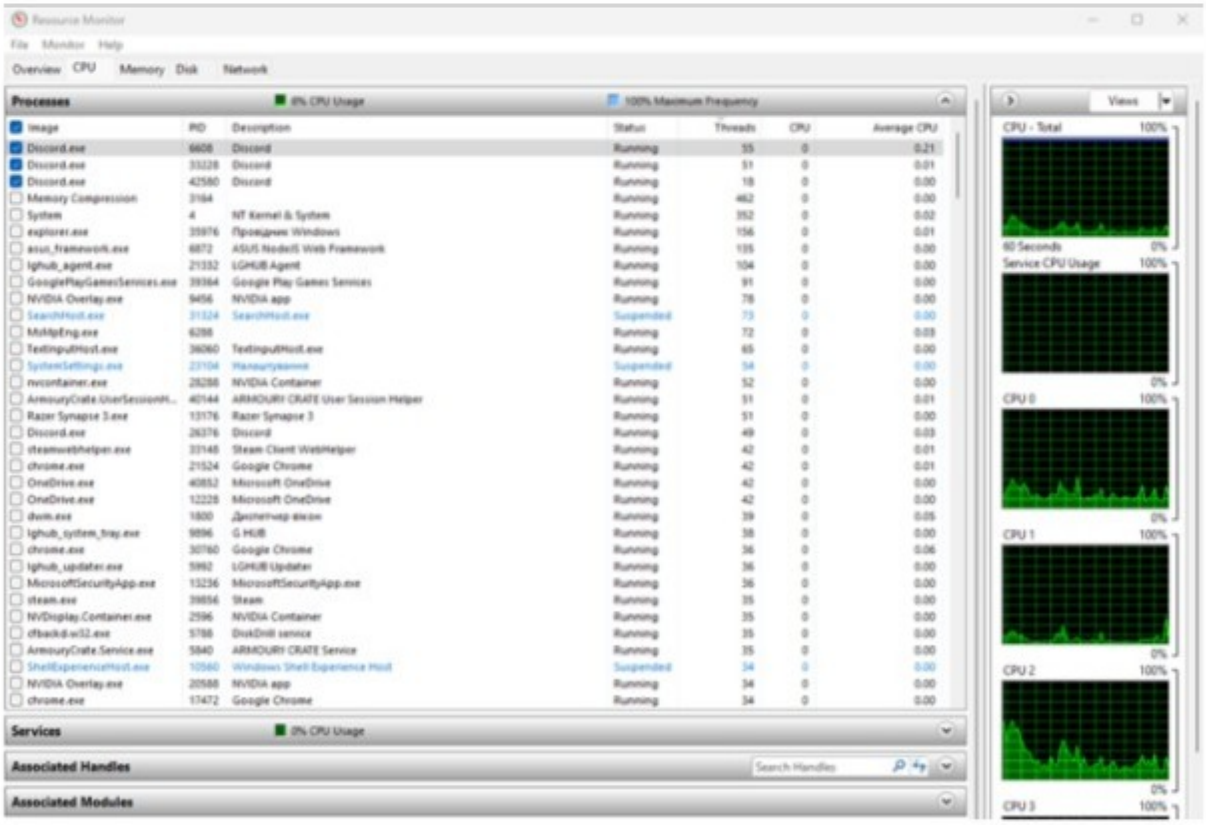


Figure 11: Discord server screen CPU consumption

As for the memory, it stays at around 621k KB for Commit, 713k KB for Working Set, 262k KB for Shareable, and 451k KB for Private.

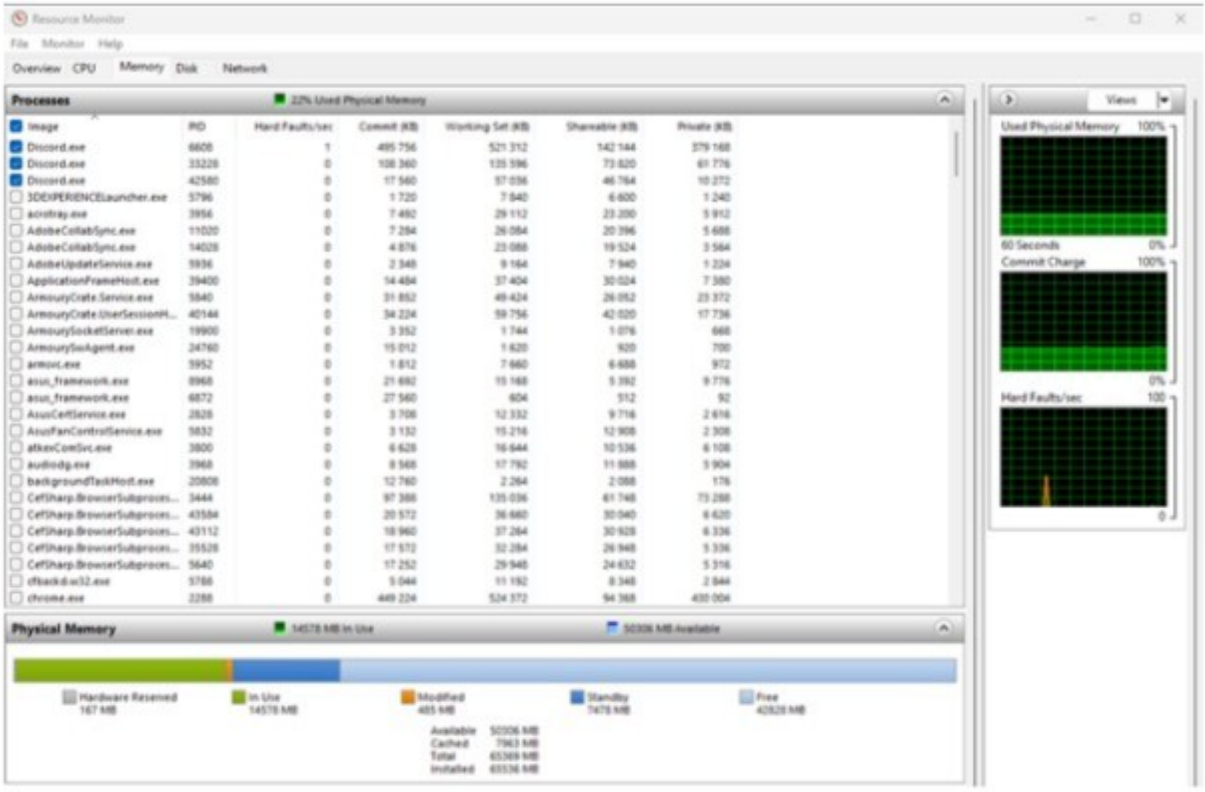


Figure 12: Discord server screen memory consumption

2. *Voice chat*



Figure 13: Discord voice chat

Joining a voice chat in the server. The CPU consumption grows a bit. From 0.22% to 0.49%.

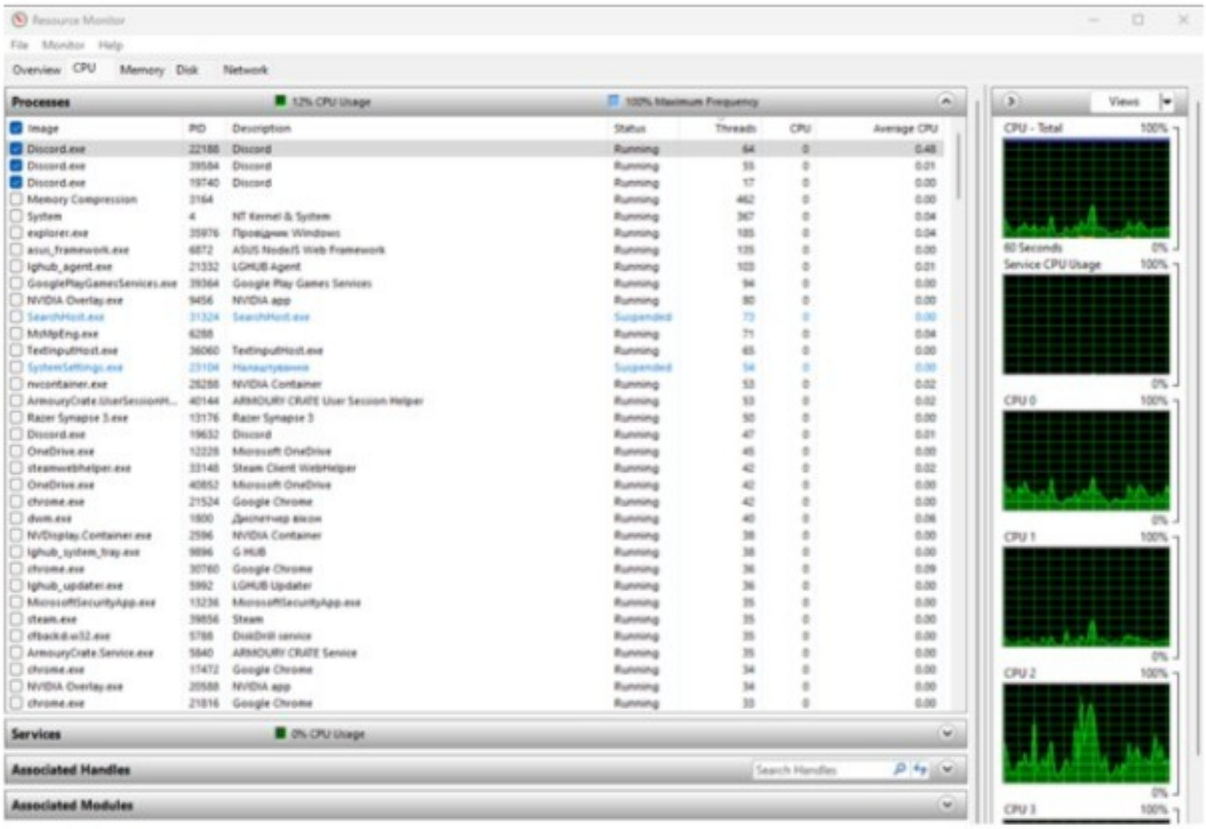


Figure 14: Discord voice chat CPU consumption

When looking at the memory consumption. It grows to 725k KB for Commit, 749k KB for Working Set, 252k KB for Shareable, and 497k KB for Private.

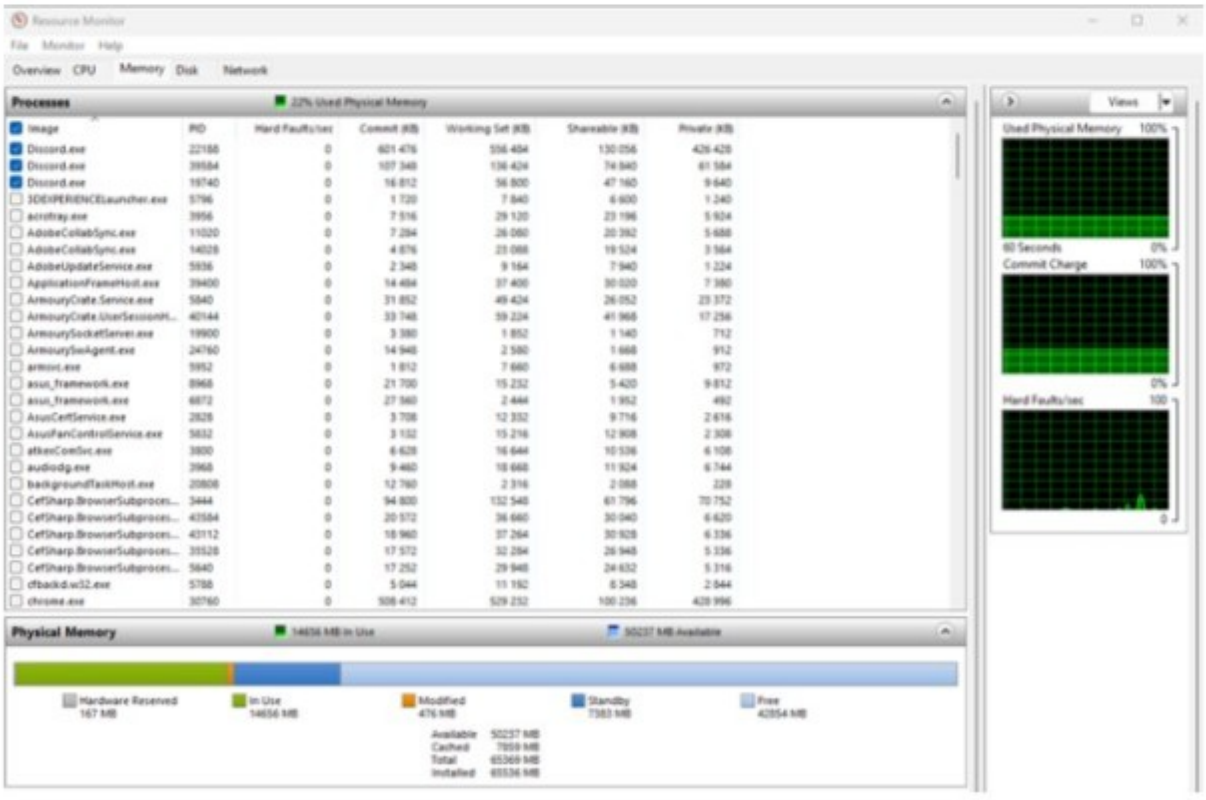


Figure 15: Discord voice chat memory consumption

3. *Screen sharing in voice chat*

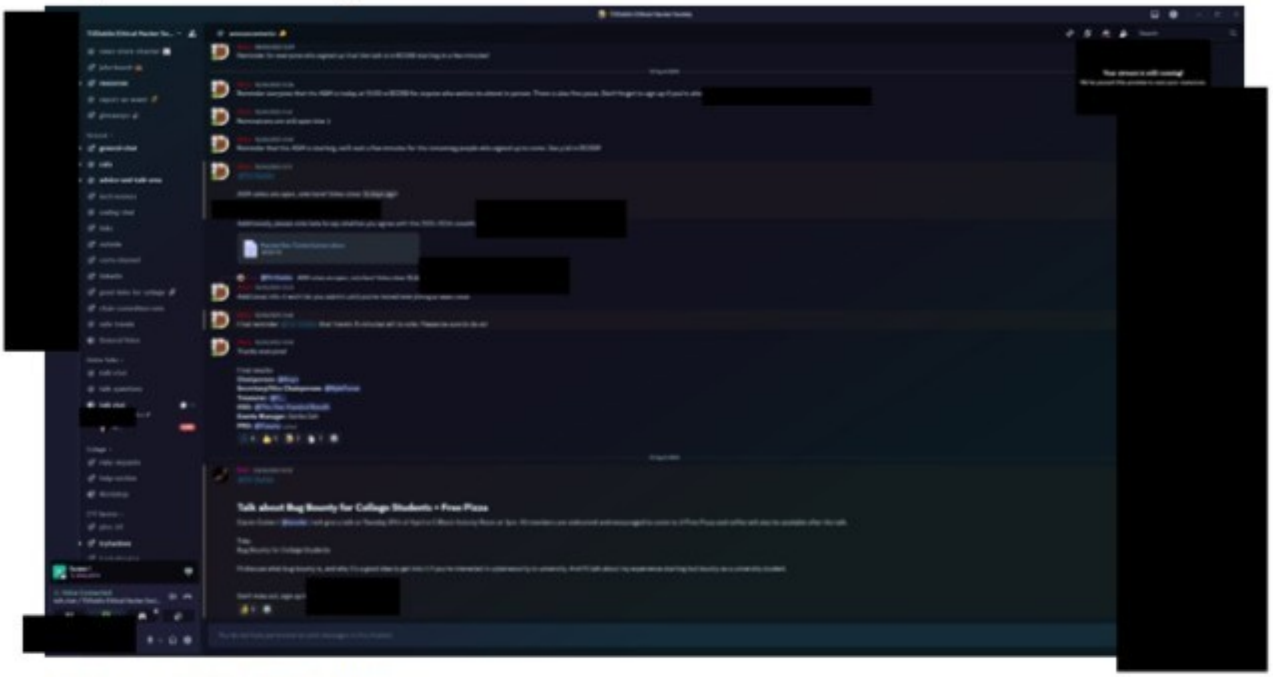


Figure 16: Discord screen sharing

While screen sharing in the voice chat on a server. CPU consumption goes up substantially. It goes from 0.49% to 1.53%.

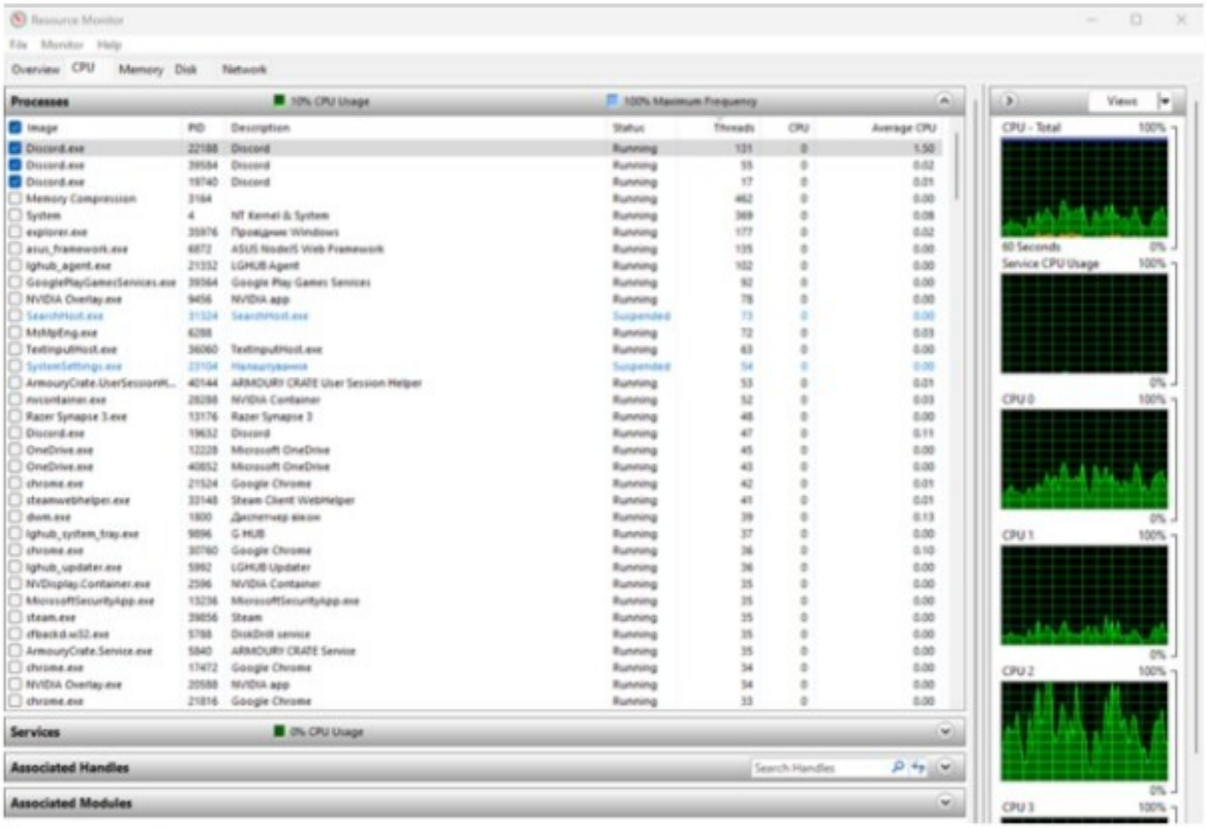


Figure 17: Discord screen sharing CPU consumption

Looking at the memory consumption, it climbs up again. It is now 1m KB for Commit, 1m KB for Working Set, 285k KB Shareable, and 752k KB for Private.

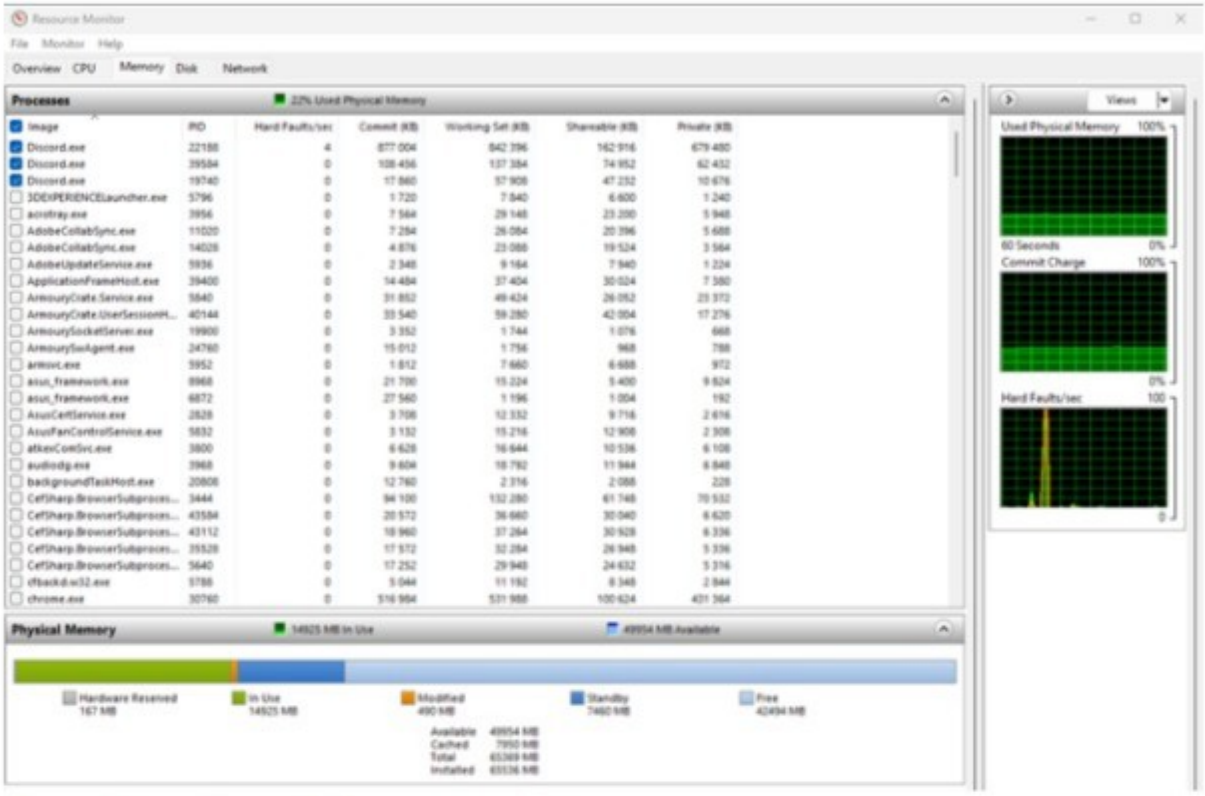


Figure 18: Discord screen sharing memory consumption

Apex Legends

Apex Legends is a 2019 battle royale-hero shooter video game developed by Respawn Entertainment and published by Electronic Arts, set in the same science fiction universe as Respawn's Titanfall series. (Wikipedia 2024b)

1. *Main menu (idle)*



Figure 19: Apex Legends main menu

When sitting in the main menu, the CPU consumption is fairly high. The average percent of CPU consumption (every 60 seconds) is 1.65%.

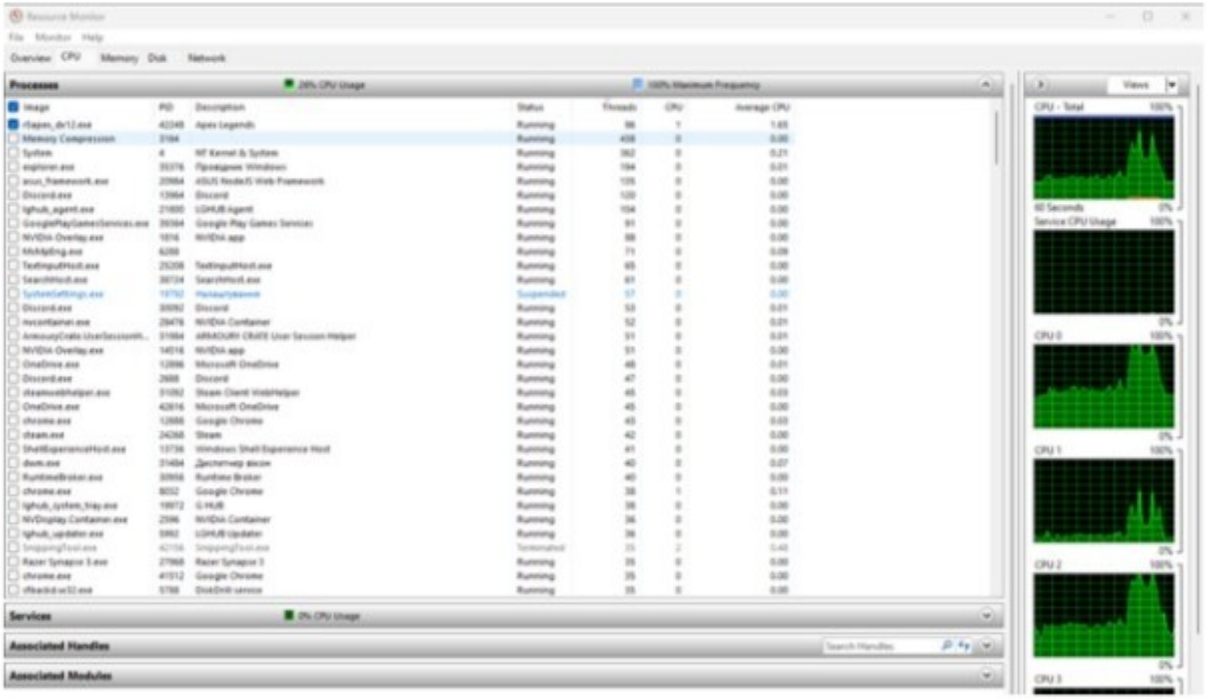


Figure 20: Apex Legends main menu CPU consumption

As for the memory, it stays high at around 9.8 million KB for Commit, 7.7 million KB for Working Set, 695k KB for Shareable, and 7 million KB for Private.

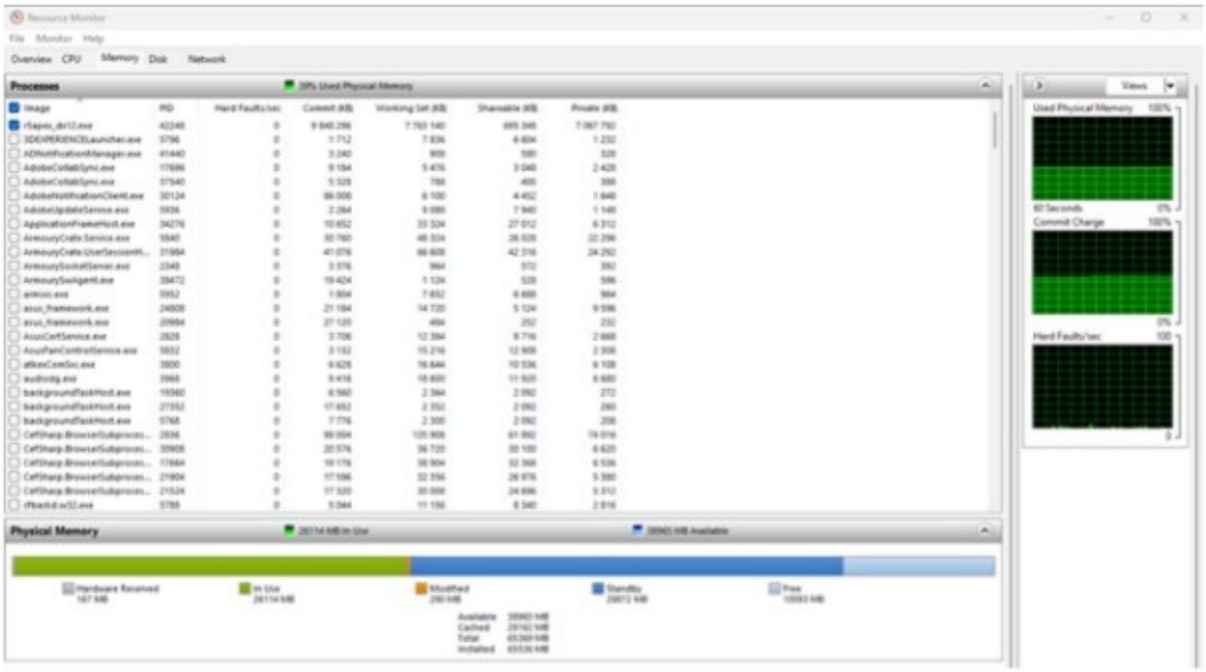


Figure 21: Apex Legends main menu memory consumption

2. Firing range



Figure 22: Apex Legends firing range

While in firing range the CPU consumption higher than in main menu. The CPU consumption raises from 1.65% to 2.67%.

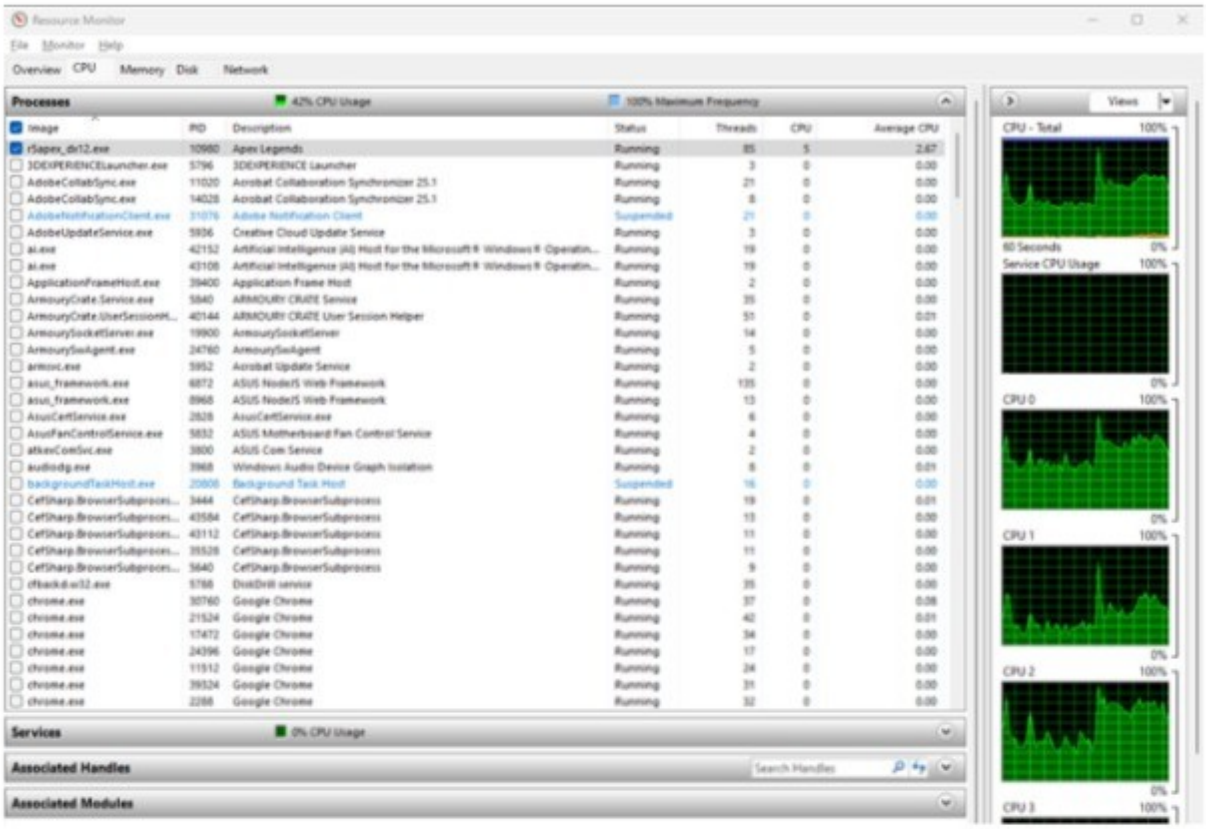


Figure 23: Apex Legends firing range CPU consumption

When looking at the memory consumption. It grows to 10.7 million KB for Commit, 6.8 million KB for Working Set, 719k KB for Shareable, and 6.1m KB for Private.

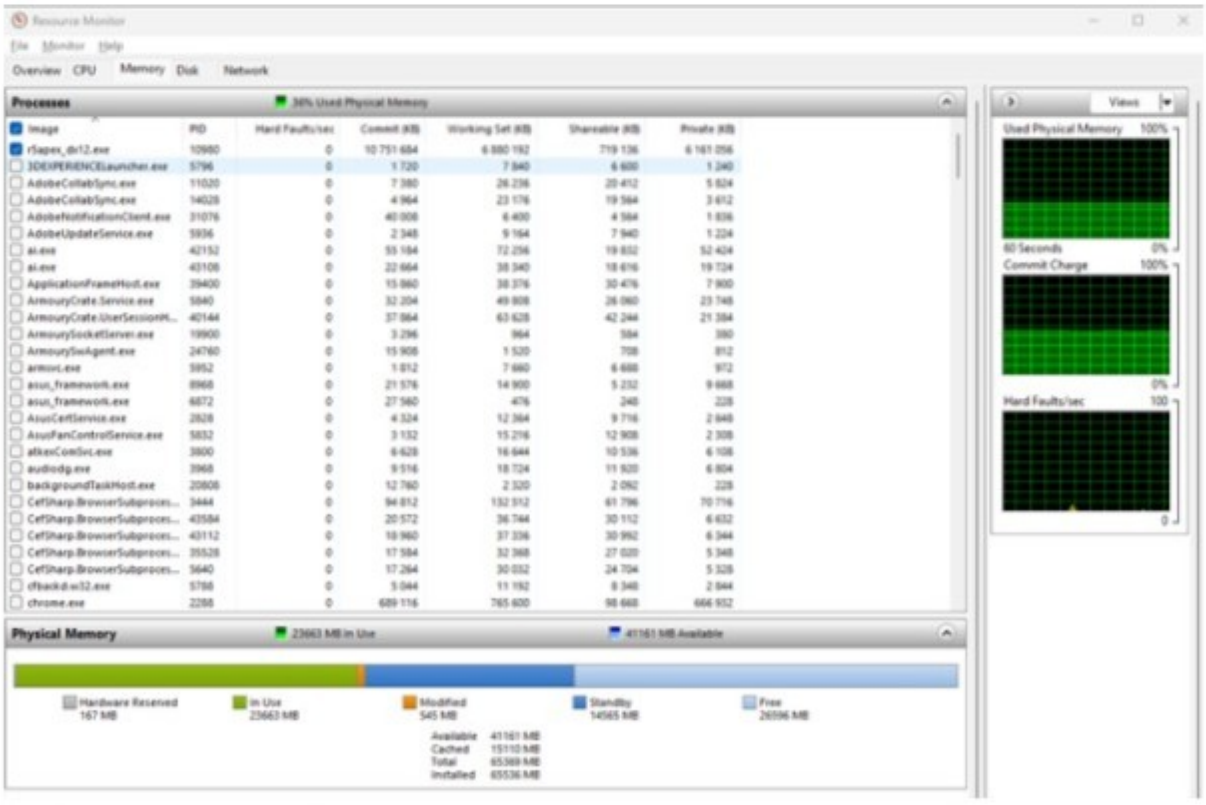


Figure 24: Apex Legends firing range memory consumption

3. *Playing a match*

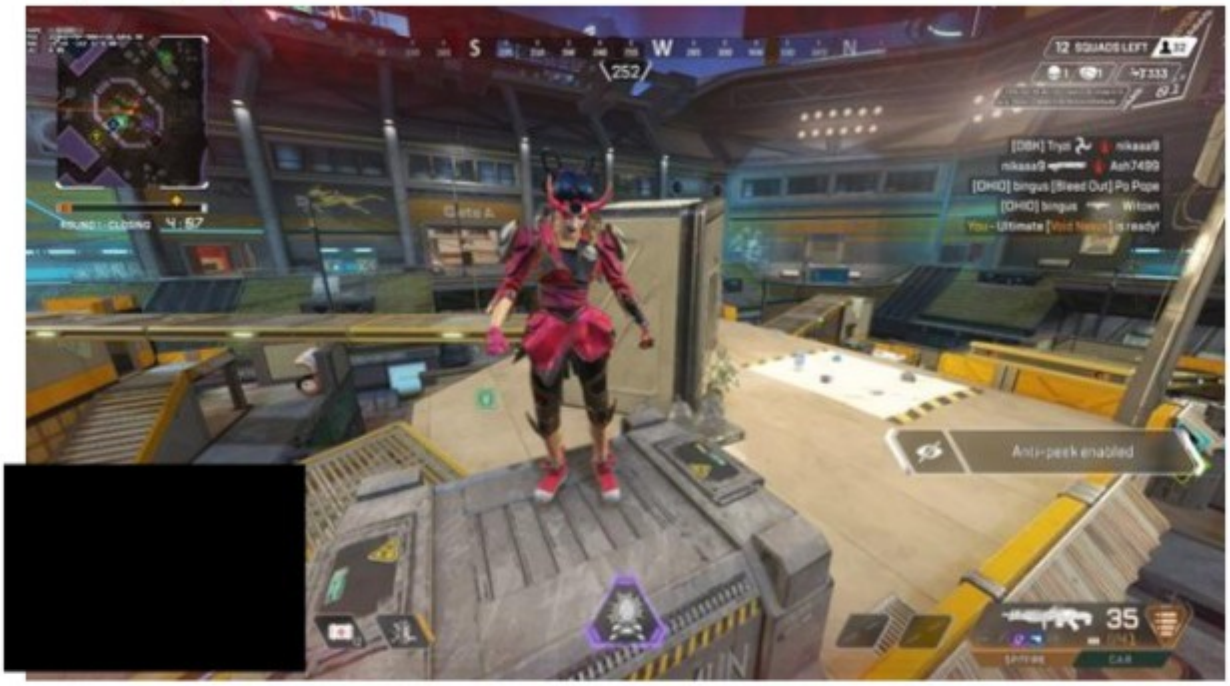


Figure 25: Apex Legends gameplay

While playing a match, the CPU consumption goes up. It goes from 2.67% to 3.55%.

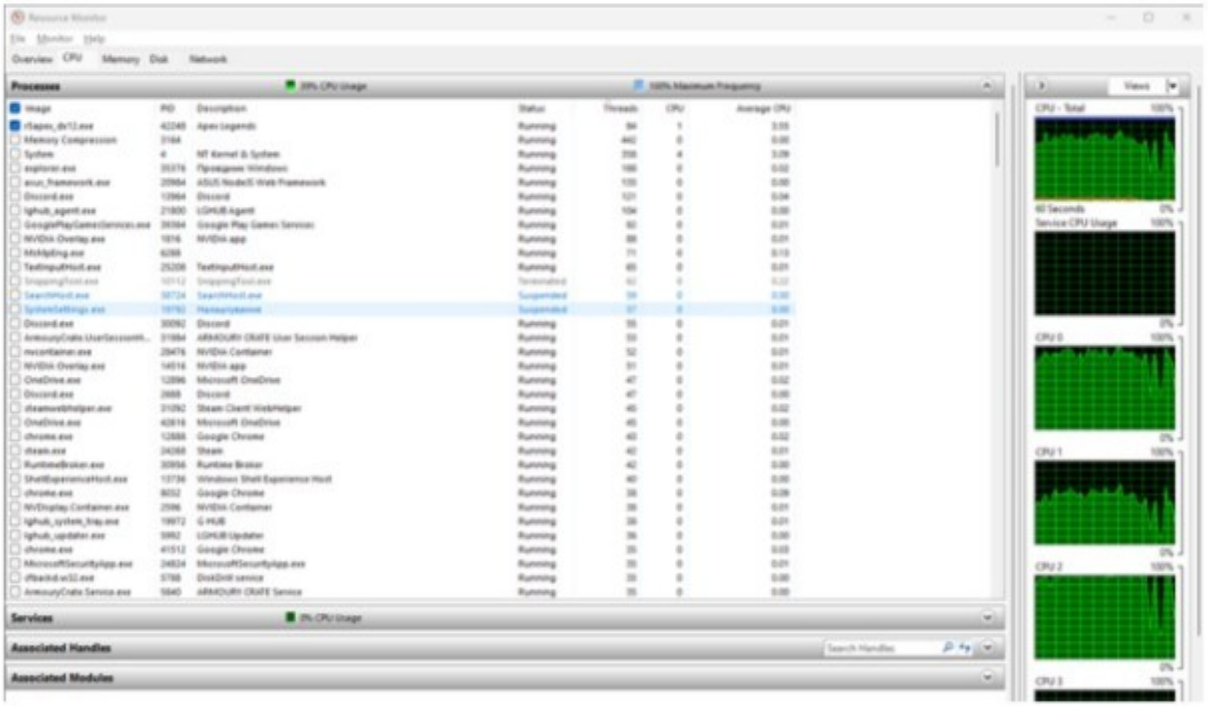


Figure 26: Apex Legends gameplay CPU consumption

Looking at the memory consumption, it climbs up from the main menu. It is now 12.9 million KB for Commit, 8.4 million KB for Working Set, 709k KB Shareable, and 7.7 million KB for Private.

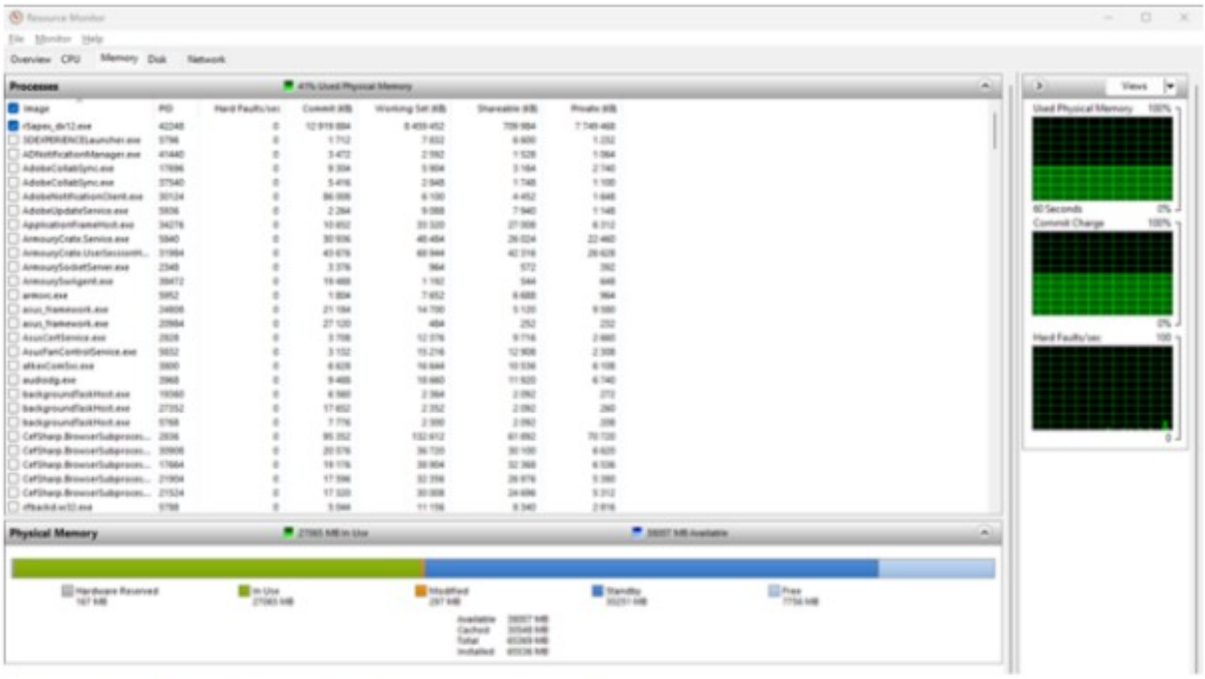


Figure 27: Apex Legends gameplay memory consumption

Conclusion

The results of the tests are summarized in this table:

Application	*Idle CPU Usage (%)*	*Active CPU Usage (%)*	*Idle Commit Memory (KB)*	*Active Commit Memory (KB)*	*Idle Working Set Memory (KB)*	*Active Working Set Memory (KB)*
osu!(Lazer)	0.01	1.12	327k	1200k	419k	748k
Discord	0.22	1.53	621k	1000k	713k	1000k
Apex Legends	1.65	3.55	9800k	12900k	7700k	8400k

Across all processes analysed in this part of the report, common patterns regarding CPU and memory utilisation arise.

Common across the tests were the CPU and memory utilisation spikes when first loading a program or substantially changing its state (e.g. loading into a game match, activating screen sharing or playing a rhythm map). This is because applications need to load required resources/assets into memory and initialise threads. Once these phases have finished, CPU usage typically stabilises or slightly decreases.

Intensive graphical programs consistently required higher CPU and memory resources. For instance, Apex Legends reached a commit memory peak of nearly 13 million KB during gameplay, reflecting the extensive loading and management of graphical assets, physics calculations, and game logic. On the other hand, software like Discord, which deals with less intensive real-time computation or graphical rendering, showed modest resource consumption even during active use.

Joseph's Processes – [redacted]

This section of the research paper focuses on music streaming, music production, the utilization of external hardware devices and the effect that each has on a computer's ability to run smoothly and efficiently.

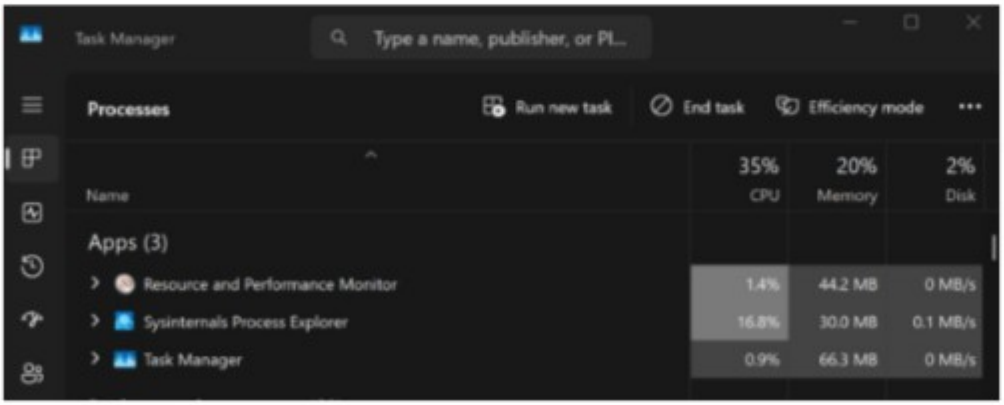
In the interest of accuracy during the research investigation, the processes were monitored independently of each other, and no additional processes (outside of the windows default) were running. The laptop was allowed time to return to rest status between tests to ensure that no background activity influenced the results.

As reference, the specs on the laptop used are as follows:
Make/Model: Microsoft Surface Book 3
Processor: Intel(R) Core i7-1065G7
Installed RAM: 32GB
Windows Version: 11 Home (24H2)

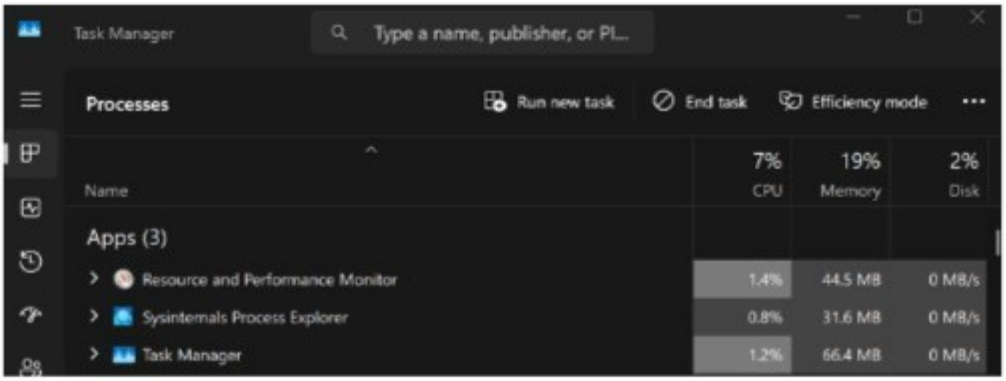
Monitoring software:
Task Manager.
Resource Monitor
Process Explorer

External device used: Boss Katana Mk1 Guitar Amplifier

It was expected that the CPU would initially spike before settling back down to a lower rate. This turned out to be the case, as the CPU was observed at 16.8% at the launch of Process Explorer, before settling back down as seen in the screenshots below.



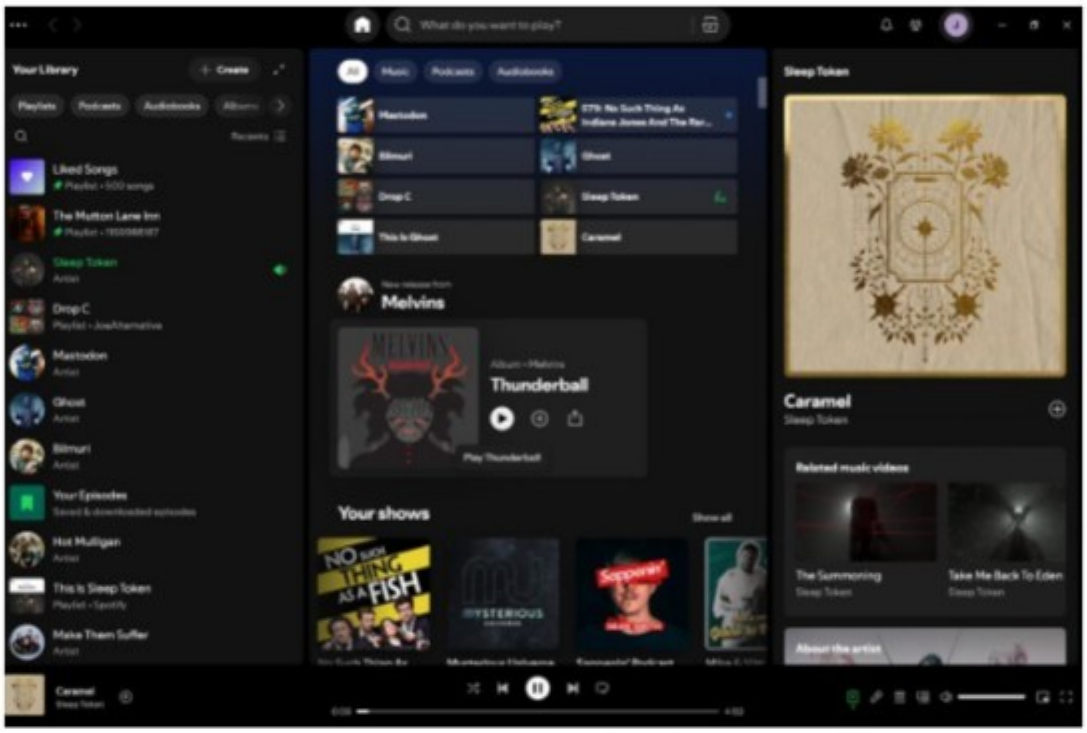
CPU Fig. 1


Task Manager
Type a name, publisher, or PI...
Processes
Run new task
End task
Efficiency mode
Name
7%
CPU
19%
Memory
2%
Disk
Apps (3)
Resource and Performance Monitor
1.4%
44.5 MB
0 MB/s
Sysinternals Process Explorer
0.8%
31.6 MB
0 MB/s
Task Manager
1.2%
66.4 MB
0 MB/s

CPU Fig. 2

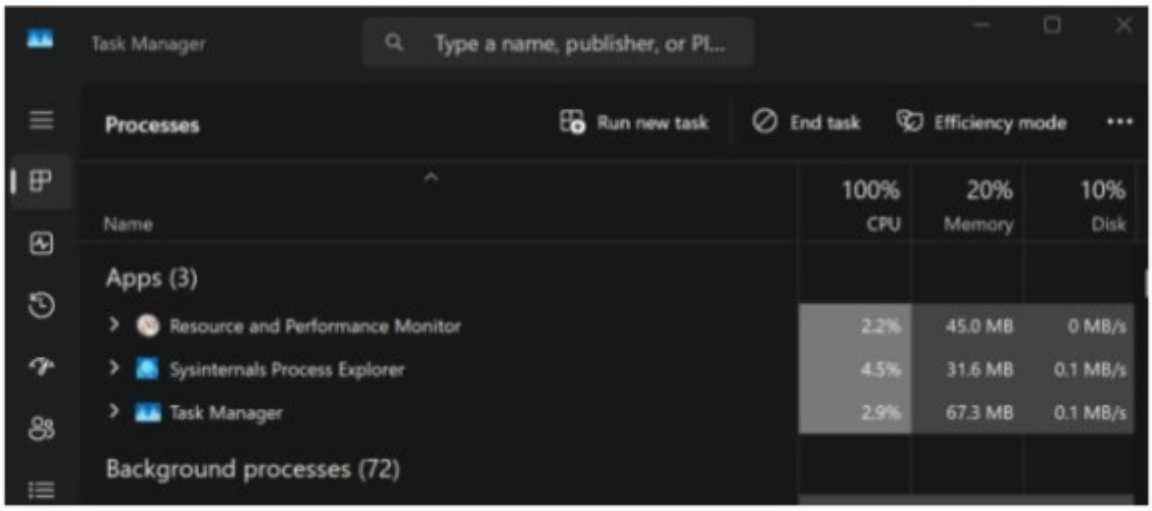
Spotify

Spotify is a music streaming service which gives the user access to the largest collection of musical artists, songs and albums as well as podcasts and audio books. It's available across all common platforms including Windows, Mac, Android, iOS as well as in browser and on video games consoles.



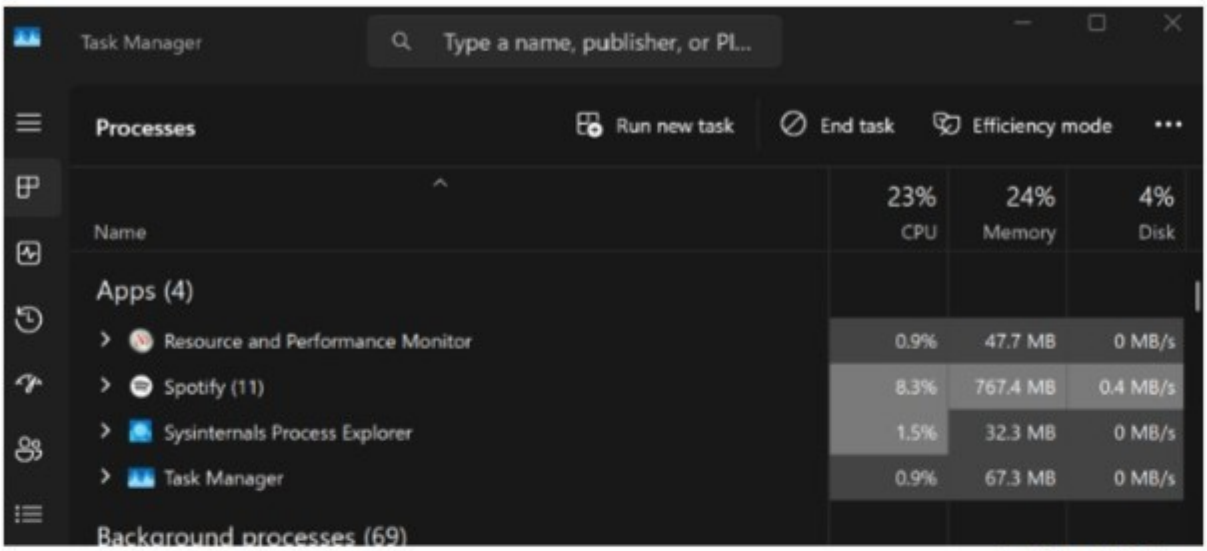
Spotify Home Screen (Windows 11 Application)

The screenshot below was captured just as Spotify was launching. Task Manager was yet to register the Spotify process, but the CPU% was captured at 100%.



Spotify - CPU Fig. 1

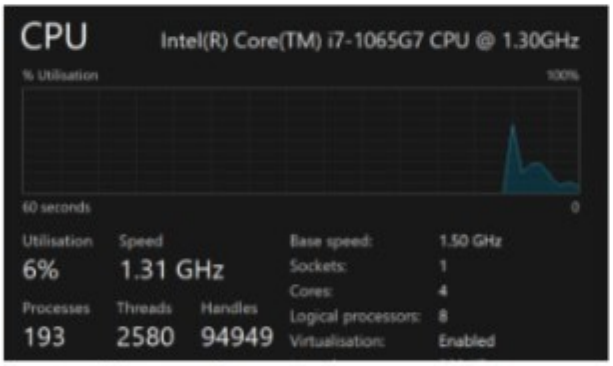
Within 1 or 2 seconds, the CPU had dropped back down to a reasonable 23% - Spotify occupying 8.3% CPU and just over 750MB of memory.



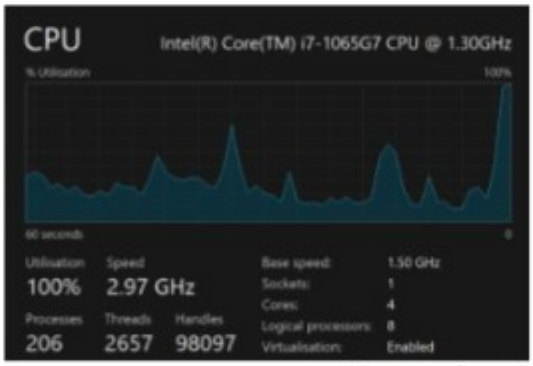
Spotify - CPU Fig. 2

Closing and relaunching Spotify, it was possible to observe the increase in processes and thread count.

This increase in thread count shows the different aspects of the program that are loading and processing when Spotify launches.



Spotify - CPU Fig. 3



Spotify - CPU Fig. 4

It was also noticed that despite Spotify being just one application, multiple instances of the Spotify image were present in the Resource monitor. This shows the different modules within the Spotify application that are required to run the software as a whole.

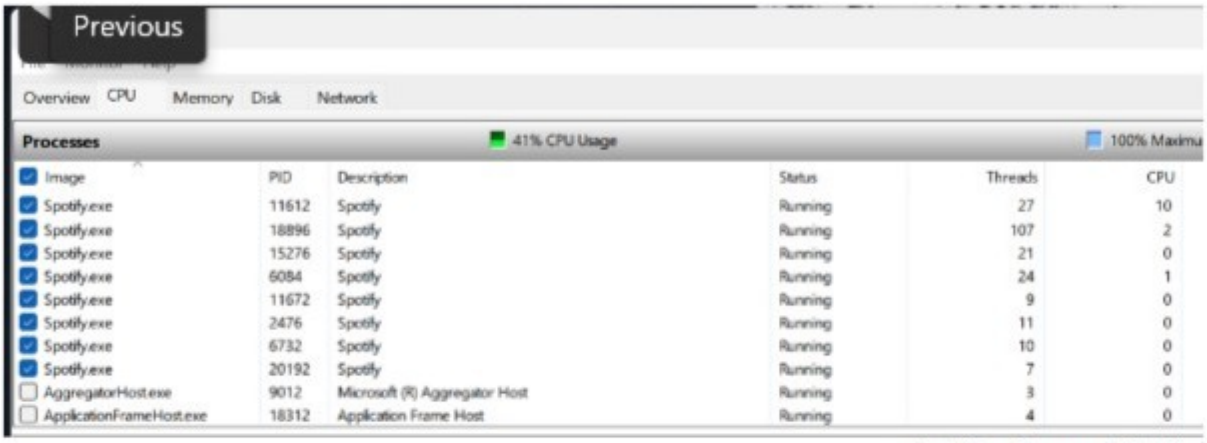


Image	PID	Description	Status	Threads	CPU
Spotify.exe	11612	Spotify	Running	27	10
Spotify.exe	18896	Spotify	Running	107	2
Spotify.exe	15276	Spotify	Running	21	0
Spotify.exe	6084	Spotify	Running	24	1
Spotify.exe	11672	Spotify	Running	9	0
Spotify.exe	2476	Spotify	Running	11	0
Spotify.exe	6732	Spotify	Running	10	0
Spotify.exe	20192	Spotify	Running	7	0
AggregatorHost.exe	9012	Microsoft (R) Aggregator Host	Running	3	0
ApplicationFrameHost.exe	18312	Application Frame Host	Running	4	0

Spotify - Resource Monitor

Boss Tone Studio.

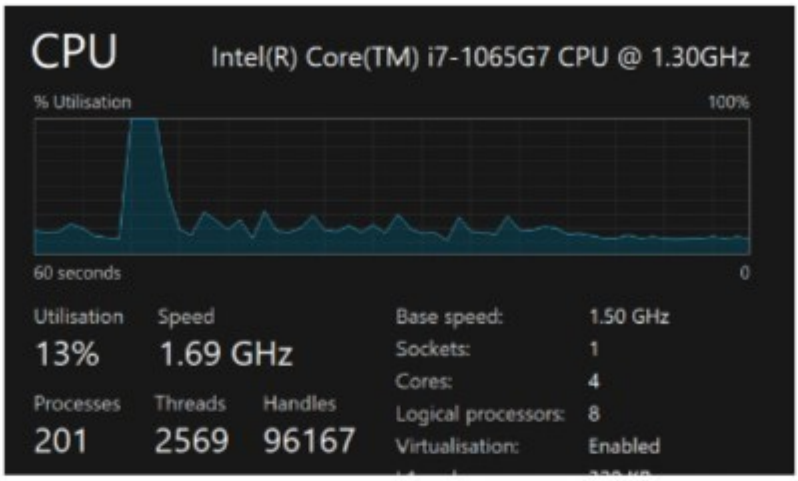
The Boss Tone Studio is free software that allows the user to control digital guitar effects stored within Boss Katana guitar amplifiers. In order to use the software, the amplifier must first be connected via USB to the computer but when connected, the user can arrange and fine tune a series of guitar effects to modify the sound of the guitar coming from the amplifier.



Boss Tone Studio

Boss Tone Studio was somewhat of a surprise having previously monitored the Spotify application. The CPU seemed far happier to manage this application. Once booted the Tone Studio software worked seamlessly, modifying the amp settings with practically no latency whatsoever.

At launch, the Page Faults showed around 45,500, increasing to approximately 50,000 with just a few slight modifications to the Amplifier settings. The screenshot below, again showing the surge of CPU processing at launch, before settling down during general use of the Tone Studio application.



Boss Tone Studio - CPU Fig. 1

Making use of Resource Monitor here, it showed the Thread Count for Tone Studio at 25 at launch, before dropping first to 24 and then 21, even with the settings modifications being sent to the amplifier via USB.



Boss Tone Studio - Resource Monitor Fig. 1



Boss Tone Studio - Resource Monitor Fig. 2



Boss Tone Studio - Resource Monitor Fig. 3

This shows that on completion of the Tone Studio boot, the threads used in startup are no longer required, lowering the thread count to only what is required for the Tone Studio to function.

Audacity

Audacity is a free, open-source, basic D.A.W (Digital Audio Workstation). It is used to record and edit audio files.
In this instance, the Boss Katana Amplifier is being used as an external audio interface, but the Tone Studio software is not running.

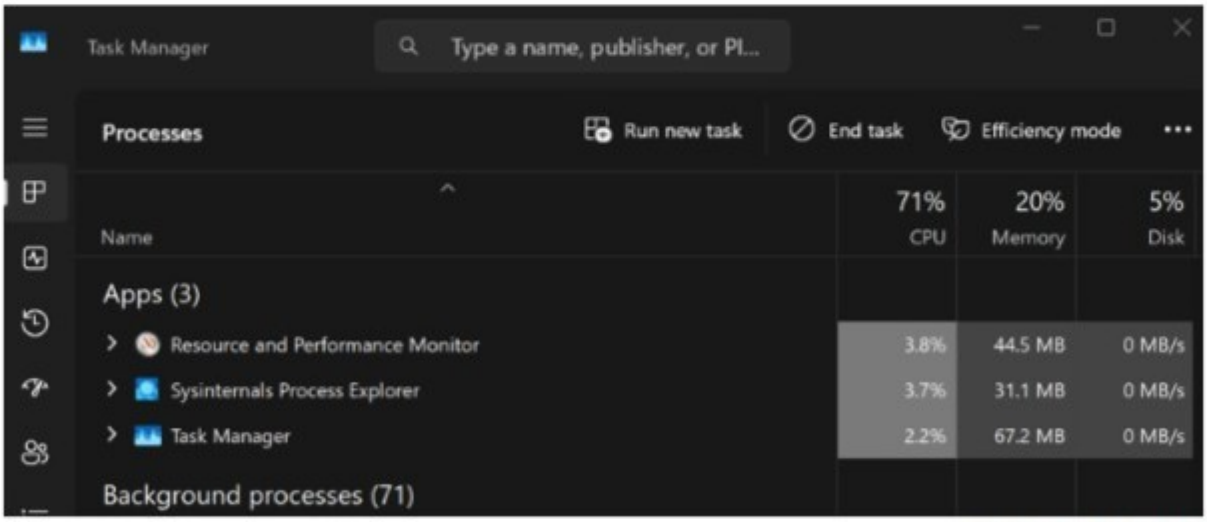


Audacity Splash Screen



Audacity Application

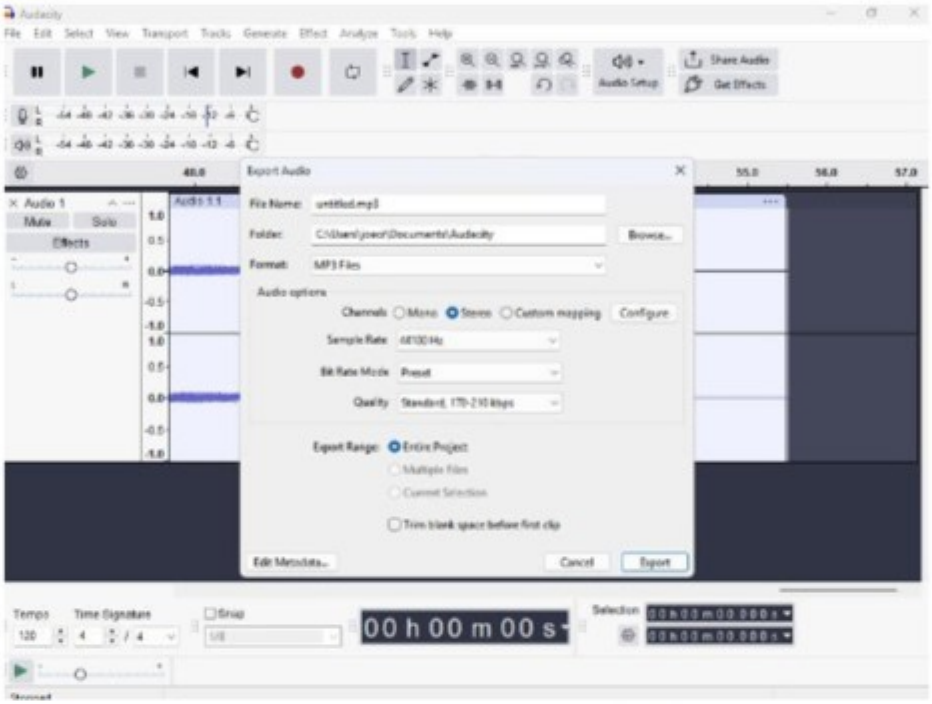
Similarly to Boss Tone Studio, Audacity showed a brief spike in CPU usage at launch, but settled quickly and was under no real stress during the research period. As above with Spotify, this image shows the CPU usage just before Task Manager registers the launch of Audacity.



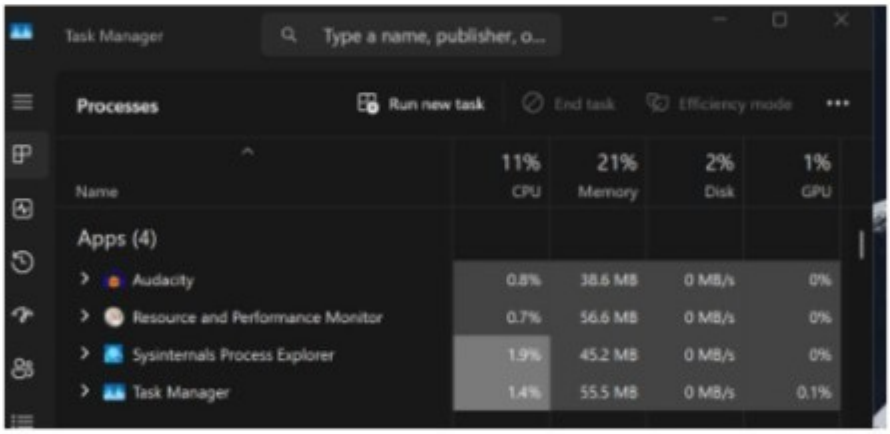
Audacity - CPU Fig. 1

What was found to be the most remarkable aspect of this investigation was after recording a few seconds of Audio and exporting an .mp3 file. The system only registered a small increase in CPU or memory usage. An increase from 0.8% to 6.6%

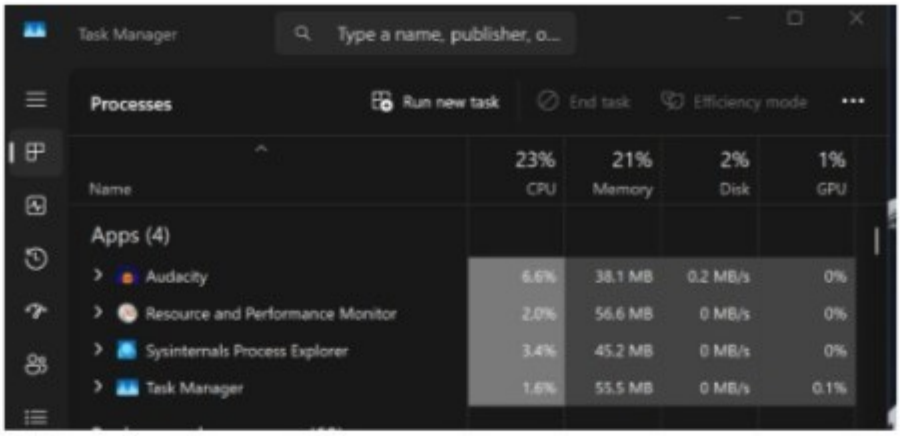
in what was expected to be perhaps the most intensive process of this research paper. In addition, the memory utilisation even slightly decreased (0.5MB) which although slight, was another interesting observation. It would indicate that the system freed cached information or buffers that were not necessary while Audacity was exporting the .mp3 file.



Audacity - Export .mp3 Fig. 1



Audacity - Export .mp3 Fig. 2



Audacity - Export .mp3 Fig. 3

It is possible to see, however, the Page Fault counter instantly rose by over 12,000. This shows the speed at which these events occur - swapping pages between the RAM and the disk space.



Audacity - Process Explorer Fig. 1



Audacity - Process Explorer Fig. 2

Notes and observations

As to be expected, the main CPU consumption occurs on launch of each application, rather than when each task within the application is performed.

It was initially surprising to see that Spotify was the main culprit when it came to CPU usage. This can be explained by even looking at the application screenshots, Spotify does have a busier User Interface, with multiple things happening simultaneously; downloading, decoding, buffering and a constant network connection to name just a few.

Audacity also was quite surprising. Despite being known as a straight-forward D.A.W, it was expected that beyond launch - recording, playing, choosing different input options, etc - it would show some strain on the CPU or memory usage. However, even exporting recorded data proved to require very little processing power which was an interesting observation.

In addition, it was noticed that each instance of capturing a screenshot also caused consistent increases in CPU usage, while triggering a number of screenshots successively caused CPU% readings to spike far more excessively than most of the processing within the applications themselves.

Despite brief spikes - which are to be expected at particular times (specifically program launch), I would say I'm quite satisfied with the performance of my laptop with how it handled these applications.

References

Reddit (2024) *osu!lazer game discussion*. Available at: https://www.reddit.com/r/osugame (Accessed: 25 April 2025).

Wikipedia (2024a) *Discord (software)*. Available at: https://en.wikipedia.org/wiki/Discord (Accessed: 25 April 2025).

Wikipedia (2024b) *Apex Legends*. Available at: https://en.wikipedia.org/wiki/Apex_Legends (Accessed: 25 April 2025).

[redacted]

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We declare that the work we are submitting for assessment by the examiner is entirely our own work, except where the author or source has been duly referenced and attributed.

We confirm that this material has not been previously submitted for a degree or any other qualification at TUD or any other institution. We further confirm that we have read and understood the Institute policy on plagiarism in assignments and examinations (3AS08.doc) and that we are not, so far as we are aware, in breach of any of these regulations.

Signed :



Date : 25/04/2025